

Financial Stimulus and Microfinance Institutions in Emerging Markets*

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Abstract

We study how microfinance institutions (MFIs) shape the allocation of financial stimulus during recessions and, through this channel, its aggregate impact. We develop a stylized model showing that private lenders may deviate from the social optimum, prioritizing inframarginal clients when they face poaching threats. This distortion is amplified when lenders are specialized in such clients. We then exploit a large-scale loan guarantee program for small firms implemented in Peru during the COVID-19 recession, combined with administrative data covering the universe of small business loans. We show that guarantees significantly reduced delinquency rates, particularly among micro-firms and those in high-contact industries, segments facing severe liquidity needs. MFIs channeled a larger share of guarantees to these high-sensitivity segments, while traditional banks prioritized inframarginal firms, consistent with our model. We extend our framework to incorporate lender specialization and heterogeneous firms and calibrate it to the data. The model shows that the program reduced defaulting debt by 30 percent relative to a counterfactual where only traditional banks distributed guarantees. Our core insight that MFI specialization shapes the aggregate impact of financial stimulus remains robust across settings where MFIs face higher distribution costs, serve riskier clients, or have weaker monitoring incentives.

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1 Introduction

Loan guarantees are a widely used policy tool for sustaining small business credit during recessions. A key feature of these programs is their distribution through private lenders, a design intended to prevent political interference. Yet, it remains unclear whether private lenders actually direct guarantees to the most sensitive, credit-constrained firms. Instead, they may prioritize inframarginal clients, such as existing large borrowers, or shift toward riskier firms when government guarantees reduce lenders' skin in the game.

In emerging markets, microfinance institutions (MFIs) play a major role in small-firm credit. Unlike traditional banks, which rely on hard-information, MFIs specialize in relationship lending. This technology allows them to serve smaller firms that traditional banks often overlook, a segment that tends to be the most sensitive to liquidity support. However, the same specialization concentrates MFIs' portfolios toward opaque borrowers, precisely those who rely on the continuous monitoring that guarantees may erode. Therefore, when designing these programs, policymakers must weigh the targeting gains from MFIs against the potential of exacerbating risk due to the lack of skin in the game inherent to loan guarantees.

In this paper, we study how lender incentives shape the allocation and aggregate impact of small business loan guarantees, and whether specialized lenders such as microfinance institutions (MFIs) can improve the allocation of these programs in developing countries.

We address these questions in the context of Reactiva Perú, a large-scale loan guarantee program implemented by the Peruvian government during the COVID-19 pandemic. This setting offers two key advantages. First, the program, equivalent to 8% of GDP, operated in a financial system where the microfinance sector is mature, accounting for over half of all small-firm lending.¹ This allows us to study the role of MFIs at scale, which is crucial

¹Peru represents a leading example of microfinance development, ranking consistently among the top markets for MFIs according to the Inter-American Development Bank. See, for example, https://graphics.eiu.com/assets/images/public/Microscope_on_Microfinance_2014/EIU-Microscope-Dec-2015.pdf.

to understanding how they shape transmission of micro-credit shocks (Breza and Kinnan 2021). Second, comprehensive regulatory reporting provides loan-level data on firm debt across traditional banks and MFIs, enabling us to measure credit balances and guarantee allocation by lender type. Together, these features allow us to identify the causal effects of loan guarantees and quantify how MFI participation shapes their allocation and aggregate effectiveness.

We use administrative data covering the universe of small-firm lending relationships in Peru from 2019 to 2021. For each lender-firm pair, we observe outstanding loan balances, delinquency, and city of loan origination. Firm-level characteristics include industry, age, and lender-reported risk rating. We merge these financial data with annual firm tax records containing sales, wage bills, and capital stock for all formally registered small firms. The granularity of our data allows us to trace which firms received guarantees and from which type of lender, directly linking lender composition to program effectiveness.

Our analysis proceeds in three steps. First, we develop a conceptual framework to illustrate the role of lender specialization. Private lenders deviate from the social optimum and this distortion is amplified when lenders specialize in large, inframarginal borrowers, a key difference between traditional banks and MFIs. Second, we estimate the causal impact of the program and show that MFIs channeled a larger share of guarantees to the most sensitive firms, while traditional banks disproportionately served inframarginal borrowers, as our framework predicts given their ex-ante specialization. Third, we build and calibrate a quantitative model with lender specialization and heterogeneous firms to quantify the optimal participation of MFIs.

In the first part of the paper, we build a conceptual framework to illustrate how lender specialization shapes the allocation of loan guarantees. We consider a continuum of heterogeneous firms, each characterized by revenue and debt obligations, which together determine net cash-flow. We model the recession as a firm-specific, unexpected liquidity shock. Firms

survive if their net cash-flow, plus any guaranteed loan received, exceeds the shock.

We define the social planner’s problem as maximizing the total value of non-defaulting debt, consistent with the program goals. Firms are ranked according to their treatment effect, defined as the increase in survival probability from receiving a guarantee, and funds are allocated to those with the highest value until the budget is exhausted.

Private lenders, in contrast, capture future profits from relationships that survive and face poaching threats when they deny guarantees. Their allocation rule therefore weighs the firm’s treatment effect and its survival probability absent the program, creating an inefficiency: lenders favor inframarginal clients over more sensitive borrowers. This distortion is amplified when lenders specialize in large firms, a key characteristic of traditional banks relative to MFIs. Thus, because the relative treatment effects in their portfolios are sharply different, the mix of lenders distributing guarantees shapes the program effectiveness beyond the standard skin in the game channel.

In the second part of the paper, we estimate the causal effects of the program using a difference-in-differences design that exploits cross-sectional variation in lenders’ guarantee uptake. We compare credit supply from high- and low-treatment lenders to the same firm before and after the program. The key identifying assumption is that, absent the program, credit supply would have evolved in parallel trends across lenders with different treatment intensities.

We provide evidence supporting our identifying assumption in three ways. First, we plot event-study graphs showing no evidence of pre-trends. Second, we control for time-varying, lender-size-dependent shocks by comparing financial institutions within the same size quartile, mitigating concerns that endogenous bank-firm matching drives the results. Third, we exploit a three-month delay in the program’s implementation as a placebo test, documenting that our treatment measure is uncorrelated with credit during the initial COVID-19 restrictions, before guarantees were distributed. We complement this evidence with a co-

ivariate balance test showing that treatment is uncorrelated with predetermined bank-level delinquency rates, leverage, ROA, and foreign-liability dependence.

We find that financial institutions with one standard deviation higher treatment increased credit supply by 10 percent while reducing non-guaranteed lending by 30 percent, suggesting partial crowding out. Aggregating our data to the firm-level, we find that one standard deviation higher treatment leads to a 14 percent expansion in firm total debt and a 36 percent contraction in non-guaranteed credit. Although the program restricted the distribution of loan guarantees to existing borrowers, the liquidity injection enabled banks to extend non-guaranteed credit to new firms. We estimate that highly treated banks increased lending to new borrowers by 42 percent.

We use a difference-in-differences instrumental variable approach to distinguish marginal from inframarginal borrowers. Using our firm-level treatment as an instrument for program participation, we find that receiving a guarantee reduces delinquency by 11 percentage points, and a 10 percent credit expansion lowers delinquency by 1.4 percentage points. These results indicate that alleviating liquidity needs was the primary channel, outweighing potential risk-shifting or weakened monitoring. Using tax records for formal firms, we document that the program operated primarily through the extensive margin. A one-standard-deviation higher treatment reduced exit probability by 1 percent in 2020 and 2 percent in 2021.²

We then identify the most sensitive borrowers using two proxies for liquidity needs during the COVID-19 recession, namely firm size and industry exposure to in-person contact (Cerdeira et al. 2023). A 10 percent credit increase reduces delinquency by 1.5 percentage points for the smallest firms (bottom 90% of the debt distribution), compared to only 0.4 percentage points for larger small firms (top 10%).³ Similarly, a 10 percent credit expansion reduces delinquency by 1.6 percentage points for firms in high-contact industries, versus 0.9

²Extensive margin is interpreted as being formal (reporting taxes) or not.

³We consider companies within the “small firm” category. All comparisons are within this segment.

percentage points for the remaining firms. Our results indicate that guarantees helped firms to alleviate the liquidity needs generated by the pandemic restrictions.

Having identified the most sensitive firms, we then compare the lending patterns of MFIs and traditional banks. We document that MFIs prioritized sensitive borrowers, allocating 65 percent of their guarantees to the smallest firms, while traditional banks allocated only 29 percent. Furthermore, 60 percent of MFIs' guarantees went to firms in high-contact industries, compared to 46 percent for traditional banks. A back-of-the-envelope calculation suggests that these allocation differences had meaningful aggregate consequences. Under the actual mix of lenders, the program reduced aggregate delinquency by 12 percentage points, compared to a counterfactual reduction of 8 percentage points if only traditional banks distributed the guarantees.

These allocation patterns closely mirror lenders' pre-recession specialization. Prior to the pandemic, traditional banks concentrated 74 percent of their small-firm lending in the largest borrowers, while MFIs allocated only 50 percent of their portfolios to this segment. The same pattern holds by industry, MFIs held 64 percent of their pre-recession portfolios in high-contact industries, versus 59 percent for traditional banks.

Lender specialization also enabled better targeting within borrower segments. We find that, among highly sensitive firms, those that received guarantees from MFIs performed better than those served by traditional banks. A 10 percent increase in credit reduced delinquency by 4 percentage points for smaller firms attached to MFIs, compared to only 0.3 percentage points for smaller firms attached to banks. Similarly, a 10 percent credit increase for high-contact firms attached to MFIs reduced delinquency by 4 percentage points, while the response for similar firms attached to banks was statistically insignificant. This evidence confirms that the composition of lenders critically shapes the allocation of financial stimulus and, through this channel, its aggregate impact.

In the third part of the paper, we extend our conceptual framework into a quantitative

model to conduct counterfactual analysis. Our framework includes heterogeneous firms and two lenders, MFIs and traditional banks, that compete à la Cournot and have different lending technologies. MFIs have a cost advantage in serving soft-information firms, while banks have an advantage with hard-information borrowers, generating distinct client distributions in equilibrium.

Upon observing a productivity signal, firms choose whether to pay a fixed cost to become formal, hard-information firms, or to remain informal, soft-information at zero cost. Once their productivity and risk are realized, firms choose capital, fully financed with debt, to operate a Cobb-Douglas technology. Our model yields different client distributions for each lender in equilibrium.

As in the conceptual framework, a liquidity shock hits firms, and the government injects guarantees distributed by lenders. The wedge between the private allocation and the social planner’s allocation arises because lenders value both treatment effects and baseline survival probability. We define the policy objective as maximizing the share of debt saved from default.

The calibrated model matches key empirical moments, including the observed specialization of lenders. MFIs clients are smaller, informal, and have less cash-in-hand, while traditional banks’ clients are larger, formal, and better positioned to survive without guarantees. As a result, MFI clients exhibit higher treatment effects, a difference that is even more pronounced among program recipients. MFI participation thus amplifies the aggregate impact by redirecting guarantees toward the most sensitive firms.

Under the observed MFI participation rate, the program saved 32 percent of debt at risk of default, compared to 22 percent in a counterfactual scenario where only traditional banks distributed guarantees. However, increasing MFI participation beyond the optimum of 57 percent dilutes effectiveness, because additional guarantees reach riskier, less responsive borrowers within MFI portfolios.

Finally, we extend our baseline model to broaden its applicability to other settings. First, we introduce distribution inefficiencies by allowing MFIs to face higher costs when administering guarantees. In this extension, the optimal MFI participation rate remains constant for low levels of inefficiency and declines when inefficiency is sufficiently high. Second, we allow MFIs to specialize in riskier borrowers by making soft-information firms inherently riskier. This reduces the optimal MFI share, as their clients become more likely to default regardless of the guarantee. Third, we consider the possibility that loan guarantees weaken monitoring incentives. Since monitoring is particularly important for soft-information firms, MFIs become less capable of targeting sensitive firms, leading to a lower optimal participation rate. Overall, our results show that the core insight that lender specialization shapes the aggregate impact of financial stimulus remains robust across these extensions.

Related Literature

This paper contributes to two primary strands of literature. First, it connects to research on government credit guarantees, a common tool for sustaining small business lending during crises (Lelarge et al. 2010; Bach 2014; Brown and Earle 2017; Mullins and Toro 2018; Bachas et al. 2021; González-Uribe and Wang 2022; Bonfim et al. 2023; Barrot et al. 2024; Core and De Marco 2024). While existing work estimates the average effects of such programs, we study the optimal design of loan guarantees to ensure financial stimulus reaches the most sensitive borrowers. We show that the composition of lenders is a critical determinant of program effectiveness.

In this line, we contribute to recent work examining how private intermediaries distribute financial stimulus (Haas-Ornelas et al. 2021; Joaquim and Netto 2022; Balyuk et al. 2021; Griffin et al. 2023). We provide novel evidence by tracing the exact allocation of guarantees across different lender types, specifically traditional banks versus MFIs, and linking

these patterns to borrower sensitivity. This allows us to quantify how the participation of specialized lenders improves the allocation and aggregate effectiveness of loan guarantees, connecting our work to the public finance literature on whether and how policies can reach marginal instead of inframarginal firms (Zwick and Mahon 2017; Criscuolo et al. 2019; Hendren and Sprung-Keyser 2020; Slattery and Zidar 2020; Kleven 2021; Cingano et al. 2025). Our findings also contribute to the literature on lender specialization (Blickle et al. 2026; Paravisini et al. 2023). We provide evidence that the pre-existing expertise of MFIs enables them to improve the allocation of a large-scale program of loan guarantees.

Second, our paper contributes to the literature on microfinance institutions in emerging markets. While early randomized controlled trials found modest real effects of microcredit (Angelucci et al. 2015; Augsburg et al. 2015; Banerjee et al. 2015b; Tarozzi et al. 2015; Attanasio et al. 2015), recent work highlights that general equilibrium forces (Kaboski and Townsend 2011; Buera et al. 2020) and large-scale interventions (Breza and Kinnan 2021; Bari et al. 2024) can generate significant aggregate impacts.

Our contribution to this literature is threefold. First, we identify a novel channel through which MFIs affect the economy: by shaping the allocation of financial stimulus during a recession. We show that MFIs are effective at distributing government guarantees to highly sensitive, smaller borrowers that traditional banks overlook, thereby crucially strengthening the program’s aggregate impact. This finding relates to the vast literature on distorted private lender incentives (La Porta et al. 2003; Claessens et al. 2008; Agarwal et al. 2016). Second, we combine detailed administrative data on the universe of MFI operations with a quasi-experimental design to trace the causal effects of a large-scale shock. Third, we integrate these findings into a theoretical framework where lender incentives determine the allocation of financial stimulus. By modeling why MFIs target smaller, sensitive firms, this framework enables us to measure their optimal participation in loan guarantee programs.

The remainder of this paper is organized as follows. Section 2 describes our data and

the institutional background, and section 3 presents our conceptual framework. Section 4 discusses our empirical approach and we report our main results in section 5. Section 6 presents our model and section 7 concludes.

2 Data and Institutional Background

2.1 Data

We combine two administrative datasets covering financial and real outcomes for the universe of small firms in Peru.

1. Credit registry data. This is loan-level data from the *Reporte Crediticio de Deudores*, provided by the Central Reserve Bank of Peru. This quarterly panel spans 2019–2021 and includes outstanding loan balances, days past due, and the city of loan origination for each firm-bank relationship. Firm-level characteristics include industry classification, lender-reported credit risk ratings, and the year of first loan origination. On the lender side, we observe a unique ID used for bank regulation purposes, the type of lender (traditional bank or microfinance institution) and balance sheet information. We use this data to estimate the effects of government guarantees on credit and delinquency rates.

2. Tax reports data. This dataset contains firm-level tax records from 2018 to 2022, including annual sales, capital stock, and employment. It also includes a unique firm tax ID, enabling precise merging with credit registry data. Geographic location and industry classification are also observed, which we use to control for local economic conditions and sector-specific trends when estimating the policy’s effects.

2.2 The Peruvian Credit Market

The Peruvian banking sector has 52 financial institutions offering business loans: 15 traditional banks and 37 microfinance institutions⁴ (MFIs). Loans are categorized into five segments based on firm sales and outstanding debt: micro-credit, small-firm credit, medium-size firm loans, large firm loans, and corporate loans. For example, micro-credit is defined as loans to firms whose total debt in the banking sector is below \$6 thousand. On the other hand, loans to corporations are defined as credit to firms whose total sales in the past two years were above \$60 million.

Table 1 provides summary statistics of lender characteristics as of December 2019. The size distribution is highly skewed: the average credit portfolio is \$1 billion, while the median is \$169 million (columns 1–2). The market is also highly concentrated, with the top 5 lenders holding 80 percent of total credit (columns 3–4). MFIs account for 13 percent of aggregate lending but play a predominant role in small business loans where they originate 68 percent of micro-credit and 47 percent of small firm credit. Crucially, we focus on micro and small firm credit, which are the segments where MFIs compete with traditional banks, and call them small business loans throughout the paper.

Lender characteristics vary substantially across credit segments, as reported in the bottom panel of Table 1. While the average lender provides \$77 million of micro-credit (around three times the median of \$28 million), the average value of corporate loans is \$1.3 billion, more than seven times the median of \$166 million. Both patterns reflect the greater right-skewness of larger loan segments.

⁴Micro-finance institutions are regulated by the Peruvian Bank Supervisor but are not classified as banks. They include saving and loan institutions, financial enterprises, and enterprises for the development of small and micro firms.

Table 1: Lender Characteristics by Loan Type

	Total Loans Mean (1)	Median (2)	Number of Banks (3)	Share Top 5 Banks (4)	Share of MFIs (5)
Total	1 106	169	52	77	12.9
<i>Loans to:</i>					
Micro-credit	77	28	42	58	68.2
Small firms	190	50	45	56	47.3
Medium-size firms	263	13	48	86	5.8
Large firms	491	8	27	87	0.3
Corporations	1 272	166	13	94	0.5

This table reports bank-level summary statistics as of December 2019. We report the mean and median of the distribution of total loans across banks for each segment of business loans. Total loans are expressed in USD million. Shares are expressed as percentages.

Table A1 reports summary statistics for firms with positive debt in December 2019. Column (1) and (2) reveal similar skewness in firm debt. The average value is around \$6 thousand, while the median is around \$500. Column (3) shows the share of delinquent firms (more than 30 days of repayment delay) equal to 12 percent in December 2019. Finally, column (4) shows that our sample is composed of around 3 million of firms, most of them in the segment of micro-credit (around 2.3 million).

2.3 Traditional Banks and Micro-Finance Institutions

MFIs expanded significantly in Peru over the past two decades, driven by broad deregulation. The industry is mature and plays a key role in the segment of small business loans nowadays. Figure 1 plots credit growth and delinquency rates for small business loans between 2018 and 2022. Panel (a) shows credit growth rates, measured as the value of credit in a given point in time relative to its value in 2017Q1. Before the recession, MFI credit grew 35 percent cumulatively, outpacing traditional banks' 20 percent growth. Panel (b) shows that

the rapid expansion of microfinance institutions was coupled with stable delinquency rates of 7%, slightly below traditional banks' delinquency.

Table A2 compares the two types of lenders. Columns (1) and (2) consider traditional banks, while columns (3) and (4) consider MFIs. Additionally, the top panel considers the whole activities of lenders, while the bottom panel only considers small business loans. Traditional banks are larger, with 20 times bigger assets and 16 times more credit. The average traditional bank serves more cities but exhibits higher geographical concentration.⁵

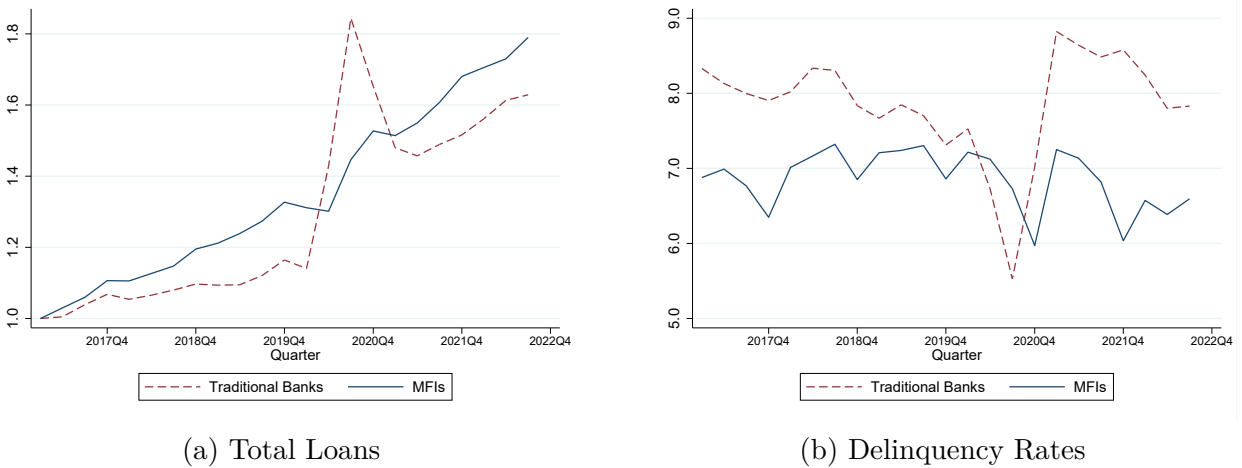
MFIs are more profitable, especially the top 10 with a ROA of 1.7 percent, reported in columns (5) and (6). The average top 10 MFI attends more markets than the average traditional bank, and also exhibits a much lower geographical concentration. Considering only small business loans, the average top 10 MFI provides \$360 million in loans, while the average traditional bank provides \$420 million.

Figure A1 maps the presence of traditional banks and MFIs across Peruvian cities as of December 2019. Around 80 percent of cities are *financial deserts*, and only 9 percent of banked locations have only traditional bank branches. In contrast, one third of banked cities are served exclusively by MFIs. While the average city served by traditional banks has 2,442 individuals per square kilometer, the average city served exclusively by MFIs has 243 individuals per square kilometer, roughly one tenth the density.⁶ Thus, MFIs play a key role in expanding geographical credit access, reaching small and opaque borrowers.

⁵We compute the geographical concentration as the Herfindahl-Hirschman index of bank portfolios across locations.

⁶The average financial deserts has 69 individuals per square kilometer, the average city with only bank branches has 1,407 individuals, and the average city with both lenders has 2,612 individuals per square kilometer.

Figure 1: Small Business Credit Growth and Delinquency by Type of Lender



This figure plots the evolution of small business credit and delinquency rates for traditional banks and microfinance institutions in the segment of small business loans. Panel (a) plots credit growth rate measured by the value of credit in a given point in time relative to the corresponding value in January 2018. Panel (b) plots delinquency rates measured by the share of outstanding debt with more than 30 days of repayment delay.

2.4 The Program of Loan Guarantees

Reactiva Perú, launched in May 2020 by the Ministry of Finance and the Central Reserve Bank, provided business loan guarantees to help firms dealing with Covid-19 restrictions. The Ministry of Finance provided the collateral, while the Central Bank provided the required liquidity. Of 52 eligible lenders, 28 participated in the program.

Guarantees covered between 80 and 98 percent of loan value (higher for smaller loans). They were allocated via first-price auctions where lenders bid on interest rates, with separate auctions for each loan type. Private lenders were in charge of screening borrowers and allocating loan guarantees. These loans had an average duration of 36 months and were granted between May and December 2020. The repayment period started 12 months after the loan was originated. Firms with poor credit rating, exhibiting more than 60 days of repayment delays in 2019, were not allowed to participate.

Table 2: Guaranteed Loans by Type of Credit

	<u>Guaranteed Loans</u>		<u>Benefited Clients</u>		<u>Guaranteed</u>
	Value	Ratio	Number	Ratio	<u>Rate</u>
	(1)	(2)	(3)	(4)	(5)
Total	15.5	29	473.1	16	91
<i>Loans to:</i>					
Micro-credit	1.2	37	319.9	14	97
Small firms	3.6	42	121.8	22	95
Medium-size firms	5.9	46	28.8	81	91
Large firms	4.5	34	2.6	82	85
Corporations	0.4	3	0.2	36	80

This table reports summary statistics of guaranteed loans in different segments of the business loan market. Column (1) reports the value of loans, in USD million, distributed by the program, and column (3) reports the number of clients, in thousand, obtaining a guaranteed loan. Columns (2) and (4) are ratios computed relative to the corresponding value as of December 2019. Column (5) shows the share of the value reported in column (1) that is guaranteed by the program.

Table 2 provides summary statistics on the program's scale for different types of credit. Column (1) shows the value of guarantees distributed, and column (2) shows the ratio of this value relative to total credit in December 2019, before the recession. Similarly, column (3) shows the total number of clients attended by the program, and column (4) shows the ratio of this number relative to the total number of borrowers in 2019. The program provided \$16 billion of guaranteed loans (29 percent of 2019 credit) and reached 473 thousand firms (16 percent of borrowers). In micro-credit, the program represented 37 percent of outstanding debt but only benefited 14 percent of borrowers, while in large firm loans, the program served 82 percent of eligible firms and represented 34 percent of outstanding credit. Guarantee rates varied by loan size, from 97 percent on micro-credit to 80 percent on loans to corporations.

Regarding lender participation, traditional banks dominated the majority of auctions at the beginning, even in the segment of small business loans. This was explained by MFIs' higher operational costs, which made their bids uncompetitive. To ensure small-firm access

to loan guarantees, the Central Bank introduced MFI-specific auctions, expanding MFI participation in the program.

We can observe in Figure 1 that, while traditional banks expanded small business loans by 50 percent peak-to-trough around the program implementation, MFIs grew only 20 percent and with a lag. Credit growth converged among both types of lenders one year after the program. Furthermore, delinquency fell sharply during the one-year grace period, following the expansion of credit. However, it rose in the post-grace period, with traditional banks reaching a 3 percentage points higher delinquency relative to MFIs.

These patterns suggest that MFIs allocated guarantees more efficiently to sensitive borrowers, a hypothesis we test rigorously in the following sections.

3 Conceptual Framework

We describe a stylized model to discuss the potential differences in lender incentives relative to the social planner’s when allocating guaranteed loans. Our framework includes heterogeneous firms with exogenous net cash-flow, which we endogenize in the quantitative analysis. The private lender chooses how to allocate guarantees over its exogenous set of clients, which we will also endogenize in the quantitative part.

3.1 Firms

There is a continuum of firms indexed by cash-in-hand y and debt obligations rb , with r denoting gross interest rate. We model the recession as an unexpected negative cash-flow shock $\nu_i \geq 0$ drawn from a distribution $\tilde{\Phi}(\nu)$. The government implements a loan guarantee program distributed by private lenders to help firms dealing with this liquidity shock. If firm i obtains a loan, it obtains a liquidity injection proportional to its current debt φb_i . Firm i survives the recession if and only if its net cash-flow $y_i - rb_i$ plus the amount received from

the program is bigger than the liquidity shock:

$$y_i - rb_i + \ell_i \varphi b_i \geq \nu_i$$

where ℓ_i is a dummy equal to one if firm i receives a loan guarantee. Then, we define the treatment effect T_i as the impact of receiving a guaranteed loan on the probability of survival:

$$T_i = \tilde{\Phi}(y_i - rb_i + \varphi b_i) - \tilde{\Phi}(y_i - rb_i) = \Phi_i(\varphi) - \Phi_i(0)$$

where $\Phi_i(z) \equiv \tilde{\Phi}(y_i - rb_i + zb_i)$. We assume that, absent the shock, all firms are solvent, i.e., $y_i - rb_i > 0$.

The first implication of our conceptual framework is that T_i depends non-linearly on the firm's net cash-flow $y_i - rb_i$. At the lowest values, treatment effects are close to zero because firms are likely to exit even with a guaranteed loan. At intermediate values, the treatment effect is maximized. At the highest values, treatment effects are again close to zero because firms would survive regardless of the program. Treatment effects also depend on the severity of the recession shock. In a deep recession, only firms with high net cash-flow can survive with assistance, while those with low net cash-flow have a survival probability near zero even if they receive guarantees.

3.2 Social Planner's Allocation

We define the social planner's optimal allocation as the allocation that maximizes the increase in non-defaulting debt attributable to the program.

$$\max_{\ell(y,b) \in \{0,1\}} \int \ell T b \, dG(y, b) \quad \text{subject to:} \quad \int \ell \varphi b \, dG(y, b) \leq M$$

where M represents the total value of guaranteed loans.

The resulting allocation is the constrained first-best, i.e., the allocation implemented by a social planner who faces the same capacity constraints as private lenders, limited by the total value of guarantees M and the individual debt increase φ among program recipients. The social planner implements this solution by ranking firms based on treatment effect T_i and distributing guarantees from the highest value until the total amount M is distributed.

3.3 Private Lenders

We consider a representative bank serving an exogenous pool of clients characterized by (y, rb) . Following Joaquim and Netto 2022, we assume that our representative bank faces poaching threats over borrowers that do not get guarantees. For simplicity, we assume that poaching threats come from an outside lender.

If the bank grants a guarantee, two outcomes are possible. If the firm survives, it repays its debt and the bank preserves the relationship, valued at $\psi_F b$ (the present value of future rents from the relationship). If the firm fails, the bank recovers a fraction δ of outstanding debt b . If the bank does not grant a guarantee, the two scenarios are the following. If the firm survives, the bank is repaid but retains the lending relationship with probability $1 - \psi_C$ (reflecting poaching risk). If the firm fails, the bank recovers a fraction δ of outstanding debt.

Thus, the bank expected profit from client i is:

$$\begin{aligned} \Pi_i = & \ell_i \{ \Phi_i(\varphi) (r + \psi_F) + (1 - \Phi_i(\varphi)) \delta \} b_i \\ & + (1 - \ell_i) \{ \Phi_i(0) (r + (1 - \psi_C) \psi_F) + (1 - \Phi_i(0)) \delta \} b_i \end{aligned} \tag{1}$$

This expression simplifies to $\Pi_i = \ell_i \Omega_i b_i + \Theta_i b_i$, with $\Omega_i \equiv T_i (r + \psi_F - \delta) + \Phi_i(0) \psi_C \psi_F$.

Then, the representative bank solves:

$$\max_{\ell(y,b) \in \{0,1\}} \int \ell [T(r + \psi_F - \delta) + \Phi(0)\psi_C\psi_F] b dG(y, b) \quad \text{subject to:} \quad \int \ell \varphi b dG(y, b) \leq M$$

3.4 Loan Guarantee Misallocation

The second implication of our framework is that private lenders weigh firm sensitivity T_i against a poaching premium $\Phi_i(0)\psi_C\psi_F$, which increases with the firm's survival probability absent the program. Thus, banks tilt towards firms that would have survived anyway. This can lead to allocative inefficiencies when treatment effects do not correlate perfectly with baseline survival probabilities, a distortion we quantify in Section 6.

To illustrate the role of lender incentives qualitatively, we consider a relationship lending parameter equal to 0.1/0.04, corresponding to a 10% annual return on preserved relationships discounted at 4%. Additionally, we assume that poaching threats realize with 10% probability.⁷

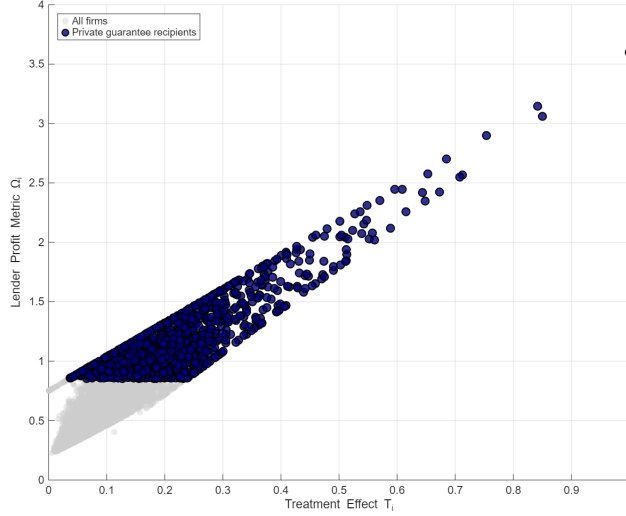
Figure 2 plots the relationship between treatment effect T_i and lender profit function Ω_i for a simulated set of borrowers, highlighting firms selected by the private lender. The allocation departs from the treatment-effect-maximizing benchmark: the private lender selects firms with low T_i but high Ω_i , as illustrated at the bottom of the selected set.

Thus, the third implication of our framework is that lender portfolios can amplify the misallocation of loan guarantees. If banks are specialized in firms with high Ω and low T , then at the margin, they will provide guarantees to firms that are likely to survive even without assistance. The degree of misallocation is therefore larger when lenders' portfolios

⁷We set the recovery rate to 10%. We model sales as a Pareto with shape parameter 1.6 over the support $[1, \infty)$ and assume that debt follows $b = 0.15y \times u$, where u is distributed as a log-normal with mean zero and variance 0.5. We can think of a revenue function $y = k^{0.3}$ with interest rate of 20%. Finally, we set the size of the program to 25% of initial debt, $\varphi = 0.8$ coinciding with the average debt increase among program recipients, and a liquidity shock distribution $\Phi(x) = (x/5)^{0.8}$ over the interval $[0, 5]$.

exhibit clients concentrated among such inframarginal borrowers. The following sections provide empirical evidence on this misallocation and quantify how the mix of lenders shapes the program’s aggregate efficiency.

Figure 2: Private Allocation of Guarantees: Lender Profit versus Treatment Effect



This figure plots the lender profit metric Ω and treatment effect T for a simulated set of borrowers. Gray dots represent the full set of borrowers and blue dots are those who receive a guarantee in the private equilibrium.

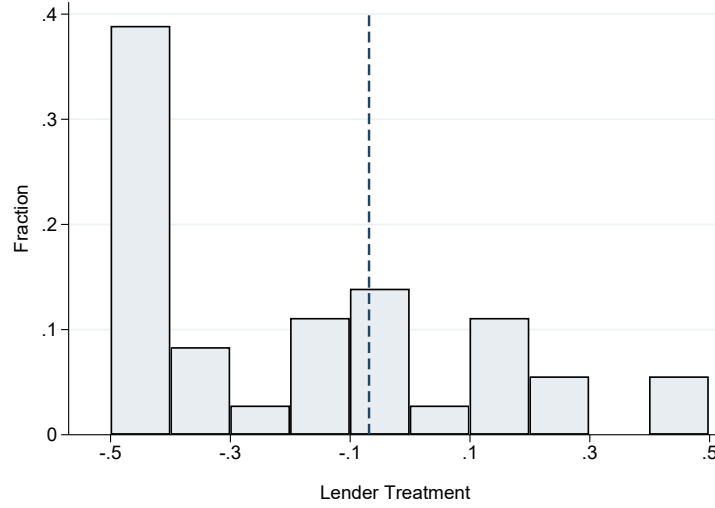
4 Empirical Framework

In this section, we describe our empirical approach to estimate the causal effects of loan guarantees and identify sensitive versus inframarginal borrowers. We exploit differences in lenders’ uptake of loan guarantees to estimate the effect of the program on credit supply. We compute the share of loan guarantees distributed by lender j , normalizing by lender j ’s pre-existing share of small business loans, as follows:

$$\text{Treatment}_j = \frac{\text{Share of Guaranteed Loans}_{j,2020} - \text{Share of Total Loans}_{j,2019}}{\text{Share of Guaranteed Loans}_{j,2020} + \text{Share of Total Loans}_{j,2019}} \times 0.5 \quad (2)$$

This continuum measure of treatment captures lender-specific reimbursement shocks (Granja et al. (2022)). Figure 3 plots the distribution of Treatment_j across lenders. We can see large heterogeneity in take-up. The dashed line indicates the median of bank treatment, weighted by pre-recession market share.

Figure 3: Distribution of Bank Treatment



This figure plots the distribution of bank treatment measured by equation 2. The dashed line indicates the weighted-by-size median treatment.

Bank-firm level specification. We identify the credit supply effect of loan guarantees using a difference-in-differences approach that compares the outstanding debt that firms hold with more treated banks relative to less treated ones, before and after the program. We estimate the following equation:

$$Y_{ijt} = \theta \times \text{Treatment}_j \times \text{Post}_t + \delta_{ij} + \delta_{it} + u_{ijt} \quad (3)$$

Figure A2 splits financial institutions into high- and low-treatment groups based on median exposure and plots the evolution of credit and delinquency rates. Panel (a) shows that treatment was uncorrelated with aggregate credit growth before the program, while high

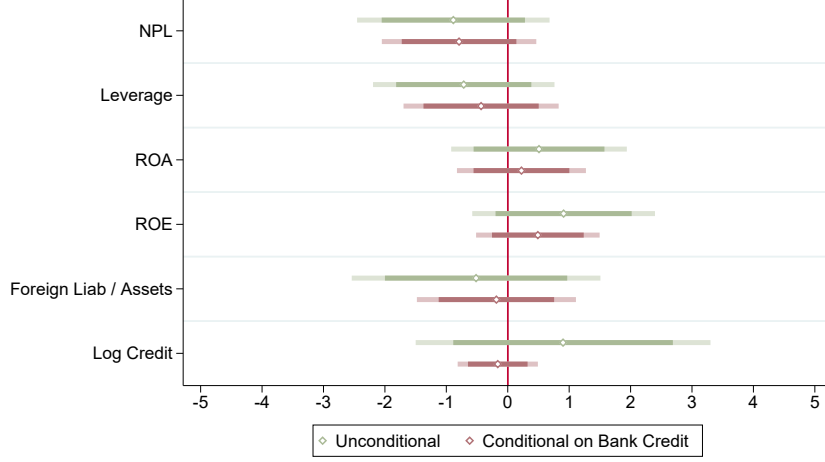
treatment predicts a rapid and persistent expansion of credit thereafter.

Panel (b) shows that treatment is also uncorrelated with delinquency before the program. Finally, while highly treated MFIs exhibit a strong and persistent reduction in delinquency rates, traditional banks' delinquency remains constant during the recession and deteriorates after the grace period. The 3 percentage point difference in delinquency among highly treated MFIs and traditional banks after the recession is economically meaningful, as small business aggregate delinquency was around 7 percent before the recession.

where Y_{ijt} denotes the balance of total loans (in logs) that firm i has with lender j in period t , and Treatment_j is the standardized treatment defined by equation (2). We include firm-bank fixed effects δ_{ij} to control for match-specific time-invariant characteristics such as lender specialization in a given industry. δ_{it} denote firm-by-period fixed effects and remove any time-varying shock at the firm level. Our parameter of interest θ measures the impact of one standard-deviation higher treatment on bank lending to the same firm. Standard errors are clustered at the bank level.

Our identifying assumption is that absent the program, credit provided by more and less treated banks would have followed parallel trends. A remaining concern, even absent pre-trends, is that larger lenders may systematically serve larger firms better positioned to deal with COVID-19 restrictions. We address this by including time-varying fixed effects for bank and firm size-bins $\delta_{g(i),g(b),t}$, which allow us to compare credit obtained from more versus less treated banks within the same size group.

Figure 4: Balance of covariates



Note: This figure reports the results of a covariate balance test. It shows coefficient estimates along with 95% confidence intervals (darker bars) and 99% confidence intervals (lighter bars) for the impact of a one standard deviation increase in bank treatment across different variables. All variables are standardized to have zero mean and a standard deviation of one. The conditional regressions include fixed effects for each bank size quartile.

While our identification strategy does not require that more and less treated banks be similar in levels, such similarity makes the parallel trend assumption more plausible. We conduct a covariate balance test regressing multiple bank-level pre-determined characteristics, separately, on our measure of treatment. Figure 4 shows the estimated coefficients and corresponding confidence intervals indicating that treatment is statistically uncorrelated with the share of non-performing loans, leverage ratio, ROA and ROE, and foreign liabilities.

Firm level specification. We aggregate our dataset at the firm level to estimate the impact of the program on small firm performance. To do so, we construct the following measure of treatment:

$$\text{Treatment}_i = \sum_j \frac{L_{ij}}{L_i} \times \text{Treatment}_j \quad (4)$$

where L_{ij} denotes the outstanding debt that firm i holds with lender j in December 2019 and Treatment_j is defined in equation (2). Then we quantify the impact of the program on

small firm performance by estimating the following equation:

$$Y_{it} = \beta \times \text{Treatment}_i \times \text{Post}_t + \delta_i + \delta_{x(i)t} + u_{it} \quad (5)$$

where Y_{it} denotes the balance of credit (in logs) or the delinquency rate of firm i in period t . We include firm fixed effects δ_i to absorb time-invariant heterogeneity. $\delta_{x(i)t}$ denotes time-varying fixed effects for the vector $x(i)$ of firm characteristics including city, industry, risk category, age-bin, and size-bin based on pre-Covid debt, absorbing demand shocks at each of these levels. Standard errors are clustered at the main-lender level.

5 Results

We organize our empirical results in three blocks. First, we estimate the effects of the program on credit supply. Second, we analyze the response of firm performance. Finally, we conduct a heterogeneity analysis to identify the most sensitive firms and study how different lenders allocate guarantees.

5.1 Loan Guarantees and Credit Supply

We start by estimating the effect of the program on credit supply. We estimate equation (3) using the log of total loans as the dependent variable. Our results are reported in columns 1 to 3 of Table 3. We find that one standard deviation higher treatment leads to a 10% increase in credit supply in our benchmark specification reported in column 2. Our results are robust to using different sets of fixed effects as reported in columns 1 and 3.

Table 3: Effect of Loan Guarantees on Bank Credit Supply

	<u>Total Loans</u>			<u>Non-Guaranteed Loans</u>		
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment _j × Post _t	0.163*** (0.050)	0.095** (0.047)	0.088*** (0.019)	-0.501*** (0.148)	-0.300*** (0.110)	-0.133*** (0.043)
Observations	37.4M	22.1M	22.1M	36.8M	21.5M	21.5M
Fixed Effects						
Bank-Firm	Yes	Yes	Yes	Yes	Yes	Yes
Quarter	Yes	–	–	Yes	–	–
Firm × Quarter	No	Yes	Yes	No	Yes	Yes
$g(i) \times g(b) \times \text{Quarter}$	No	No	Yes	No	No	Yes

This table shows the effect of the program on the balance of total loans and non-Covid-19 loans at the bank-firm level. Treatment is standardized. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively. Standard errors are clustered at the bank level. Observations are expressed in millions.

Figure 5 plots event study graphs for the response of credit supply, showing quarterly treatment effects normalized to zero in the quarter before implementation. Treatment had null effects before the policy, consistent with our identifying assumption.

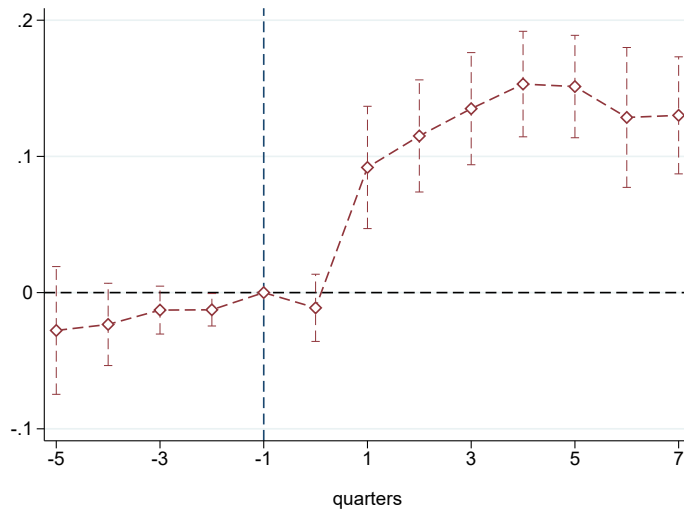
Moreover, our estimated coefficients are statistically indistinguishable from zero up to the first quarter of the policy, two quarters into the recession and before any meaningful volume of guarantees was distributed. Even though our identifying assumption is not testable by definition, this evidence provides critical support to our identification because it indicates that treatment was orthogonal to factors affecting firm credit demand when the shutdowns were already in place for two quarters. The balance of loans experience a significant and persistent increase since the third quarter of 2020.

We next examine whether government-backed guarantees crowd out normal bank activity (Stiglitz (1993), La Porta, Lopez-de-Silanes, and Shleifer (2002), Ru (2018)). We estimate equation (3) using the log of normal loans, those that are not collateralized by the government, as our dependent variable. We report our results in columns 4 to 6 of Table 3.

We estimate that one standard deviation higher treatment leads to a decline of 30% in the supply of normal loans in our benchmark specification reported in column 5.

Figure A3 plots the event study graph associated with our benchmark specification showing no evidence of pre-trends. The balance of normal loans exhibit a steady decline after the program, indicating that loan guarantees partially crowded out the normal activity of banks. Thus, the program of guarantees fully explains the expansion of total credit during the recession.

Figure 5: Effect of the Program on Total Credit



This figure plots the quarterly effects of the program on total credit and non-Covid loans at the bank-firm level. The dependent variable is in logs. The program is implemented in the second quarter of 2020. Each dot is the coefficient on the interaction of treatment and quarter fixed effects. We normalize the treatment effect at the quarter right before the implementation of the program to be equal to zero. Confidence intervals at 95%.

5.2 Loan Guarantees and Small Firm Performance

To study the impact of loan guarantees on small firm performance, we aggregate our data at the firm level and calculate a corresponding measure of treatment as described in equation (4), which captures how well connected small firms were with more treated banks before the

program. While the program led to an expansion of credit provided by highly treated banks, this does not mean better connected firms will receive more credit. If lending relationships were fully flexible, firms would easily switch to highly treated lenders and obtain guaranteed loans regardless of their pre-recession lender. Otherwise, better connected firms would receive more credit relative to less well connected ones, reflecting the stickiness of lending relationships.

We estimate equation (5) using total loans as our dependent variable. Table 4 (columns 1–3) shows that firms with one standard deviation higher treatment experience a 14% increase in total loans after the program.⁸ We report quarterly treatment effects in panel (a) of Figure A4 showing no pre-trends. Similar to our bank-firm estimation, our treatment effects are statistically null up to the first quarter of the program, where COVID-19 shutdowns were in place for two periods but only a negligible part of guarantees was distributed. We find that better connected firms have more credit, and this effect is persistent. Our results indicate that lending relationships play a key role in shaping the ability of firms to obtain guaranteed loans.

We report the response of non-guaranteed loans in columns 4 to 6 of Table 4. Firms with one standard deviation higher treatment have a 36% lower balance of non-guaranteed loans relative to less well connected firms. Panel (b) of Figure A4 reports quarterly treatment effects, showing no evidence of pre-trends.

We now explore the response of delinquency rates, our measure of small firm performance. We define delinquency as an indicator variable equal to one if the firm experiences more than 30 days of repayment delay in a given quarter. Our results are reported in columns 1 to 3 of Table 5. We find that highly treated firms perform better after the program. One standard deviation higher treatment reduces the probability of experiencing repayment delays by 2

⁸Since we cannot include firm-time fixed effects in this specification, the corresponding bank-firm level estimation is reported in column (1) of Table 3. Thus, a smaller coefficient in our firm-level specification means that firms have positive but limited ability to switch lenders.

ppts. Figure A5 plots the event-study graph associated with our benchmark specification and shows no evidence of pre-trends. Furthermore, the decline in delinquency among highly treated firms persists after the end of the grace period.

Table 4: Effect of Loan Guarantees on Small Firm Borrowing

	<u>Total Loans</u>			<u>Non-Guaranteed Loans</u>		
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment _i × Post _t	0.163*** (0.038)	0.141*** (0.022)	0.140*** (0.024)	-0.433*** (0.074)	-0.436*** (0.076)	-0.357*** (0.080)
Observations	24.5M	24.5M	12.1M	24.2M	24.2M	12.0M
Fixed Effects						
Firm	Yes	Yes	Yes	Yes	Yes	Yes
$x^{\text{Real}}(f) \times \text{Quarter}$	Yes	Yes	Yes	Yes	Yes	Yes
$x^{\text{Fin.}}(f) \times \text{Quarter}$	No	Yes	Yes	Yes	No	Yes

This table shows the effect of the program on the balance of total loans and non-Covid-19 loans at the bank-firm level. Treatment is standardized. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively. Standard errors are clustered at the bank level. Observations are expressed in millions.

We then estimate two elasticities. First, we estimate the response of firm delinquency to program participation. Second, we measure the response of delinquency to credit received from the program. These elasticities identify which firms are sensitive versus inframarginal, and allow us to quantify how reallocating guarantees across firms and lenders affects the program's aggregate effectiveness. We estimate the following IV difference-in-differences specification:

$$\begin{aligned} \text{Delinquency}_{it} &= \gamma \times X_{it} + \delta_i + \delta_{x(i)t} + u_{it} \\ X_{it} &= \beta \times \text{Treatment}_i \times \text{Post}_t + \delta_i + \delta_{x(i)t} + u_{it} \end{aligned} \tag{6}$$

where X_{it} can be either being a program recipient or total loans. In each case, we instrument this variable with our firm-level measure of treatment. Our coefficient of interest

γ measures the elasticity of delinquency to participating in the program or to the liquidity received from the program.

We report our results in columns 4 and 5 of Table 5. We find that program recipients are 11 percentage points less likely to exhibit repayment delays compared to non-participants. Furthermore, a 10 percent increase in credit reduces the probability of delinquency in 1.4 percentage points.

Table 5: Effect of Loan Guarantees on Small Firm Financial Performance

	OLS			IV	
	(1)	(2)	(3)	(4)	(5)
$\text{Treatment}_i \times \text{Post}_t$	-0.020*** (0.004)	-0.019*** (0.004)	-0.020*** (0.005)		
$\text{Recipient}_i \times \text{Post}_t$				-0.107*** (0.022)	
$\ln \text{Loans}_{it}$					-0.135*** (0.033)
Observations	24.6M	24.6M	12.1M	24.6M	24.6M
Fixed Effects					
Firm	Yes	Yes	Yes	Yes	Yes
$x^{\text{Real}}(f) \times \text{Quarter}$	Yes	Yes	Yes	Yes	Yes
$x^{\text{Fin.}}(f) \times \text{Quarter}$	No	Yes	Yes	Yes	Yes

This table shows the effects of being better connected to treated banks on the balance of total loans, non-Covid-19 loans, and delinquency rates at the firm level. Treatment is standardized. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively. Standard errors are clustered at the firm level.

Overall, the program expanded credit supply and improved small firm performance. Highly treated firms received more credit and were less likely to be delinquent after the program. The lower delinquency is consistent with the unprecedented demand for external financing generated by COVID-19 restrictions, which outweighs potential firm risk-shifting incentives and reduced lender screening.

5.3 Heterogeneous Effects and Loan Guarantees' Allocation

In this subsection, we estimate the heterogeneous effects of the program and study the role of MFIs in allocating loan guarantees towards more sensitive firms. First, we split firms into two groups based on their outstanding debt in 2019. We define firms in the top decile of the debt distribution as large borrowers, and those in the bottom deciles as small borrowers.⁹ Large firms account for 60 percent of small-firm debt in the pre-recession period. Then, we estimate equation (6) for each group of firms.

Table 6: Heterogeneous Effect of Loan Guarantees on Small Firm Financial Performance - Pre-Covid Debt Deciles

	Bottom Deciles (40% of Loans)			Top Decile (60% of Loans)		
	OLS	IV		OLS	IV	
	(1)	(2)	(3)	(4)	(5)	(6)
Exposure _{<i>i</i>} × Post _{<i>t</i>}	-0.020*** (0.004)			-0.008* (0.004)		
Treated _{<i>i</i>} × Post _{<i>t</i>}		-0.121*** (0.024)			-0.034* (0.019)	
ln Loans _{<i>it</i>}			-0.151*** (0.035)			-0.039* (0.020)
Observations	21.7M	21.7M	21.7M	2.9M	2.9M	2.9M
Fixed Effects						
Firm	Yes	Yes	Yes	Yes	Yes	Yes
$x^{\text{Real}}(f) \times \text{Quarter}$	Yes	Yes	Yes	Yes	Yes	Yes
$x^{\text{Fin.}}(f) \times \text{Quarter}$	Yes	Yes	Yes	Yes	Yes	Yes

This table shows the effects of being better connected to treated banks on the balance of total loans, non-Covid-19 loans, and delinquency rates at the firm level. Treatment is standardized. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively. Standard errors are clustered at the firm level.

Our key hypothesis is that liquidity needs are bigger among small firms. Since large firms typically exhibit higher levels of cash-flows and liquidity buffers, they will be more likely to

⁹It is worth noting that we split firms within the segment of small-firm credit. Thus, all firms in our sample are small.

survive the recession even without the program.¹⁰ Our estimation results are reported in columns (1) to (6) of Table 6. Consistent with our hypothesis, the elasticity of delinquency to credit among smaller firms is four times that of large borrowers. While a 10 percent increase in credit reduces the probability of delinquency by 1.5 percentage points among smaller borrowers, the response is only 0.4 percentage points among large borrowers.

Table 7: Heterogeneous Effect of Loan Guarantees on Small Firm Financial Performance - Industry Contact

	High-Contact			Low and Non-Contact		
	OLS	IV		OLS	IV	
	(1)	(2)	(3)	(4)	(5)	(6)
Exposure _{<i>i</i>} × Post _{<i>t</i>}	-0.021*** (0.003)			-0.016*** (0.005)		
Treated _{<i>i</i>} × Post _{<i>t</i>}		-0.116*** (0.020)			-0.088*** (0.025)	
ln Loans _{<i>it</i>}			-0.162*** (0.035)			-0.093*** (0.028)
Observations	16.5M	16.5M	16.5M	8.0M	8.0M	8.0M
Fixed Effects						
Firm	Yes	Yes	Yes	Yes	Yes	Yes
$x^{\text{Real}}(f) \times \text{Quarter}$	Yes	Yes	Yes	Yes	Yes	Yes
$x^{\text{Fin.}}(f) \times \text{Quarter}$	Yes	Yes	Yes	Yes	Yes	Yes

This table shows the effects of being better connected to treated banks on the balance of total loans, non-Covid-19 loans, and delinquency rates at the firm level. Treatment is standardized. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively. Standard errors are clustered at the firm level.

Second, we split firms based on industry. Following Cerda et al. 2023, we define high-contact industries as those with customer-facing interactions (e.g., retail, restaurants, transportation), and low-contact industries as those where such interactions are limited (e.g., manufacturing). Our hypothesis is that high-contact industries faced a larger liquidity shock,

¹⁰The alternative hypothesis discussed in our conceptual framework is that, in a deeper recession, large firms can be more sensitive to the program. This is because, given the size of the shock, small firms with low levels of cash will not survive even if they receive liquidity from the program.

as they were more directly affected by COVID-19 restrictions.

Our estimation results are reported in columns (1) to (6) of Table 7. Consistent with our hypothesis, the elasticity of delinquency to credit is bigger in high-contact industries. While a 10 percent increase in liquidity reduces the probability of delinquency by 1.6 percentage points in high-contact industries, the effect is 0.9 percentage points in low-contact sectors.

Having identified the most sensitive borrowers, those that are smaller and in high-contact sectors, we now study the allocation of guarantees implemented by each type of lender. The top panel of Table 8 reports the participation of small and large borrowers in the portfolios of loan guarantees and pre-recession small business loans for traditional banks and micro-finance institutions.

Table 8: Share of Credit and Loan Guarantees by Type of Bank and Firm

	Loan Guarantees		Pre-Recession Loans	
	Banks (1)	MFIs (2)	Banks (3)	MFIs (4)
Size (pre-Covid Debt):				
Small	0.29	0.65	0.26	0.50
Large	0.71	0.35	0.74	0.50
Industries:				
High-Contact	0.46	0.60	0.59	0.64
Less- and Non-Contact	0.54	0.40	0.41	0.36
Lender share:				
	0.75	0.25	0.48	0.52

This table reports the composition of MFI and traditional bank portfolios. Columns (1) and (2) show the distribution of loan guarantees across firm types, while columns (3) and (4) show the distribution of pre-recession loans as of December 2019. Small firms are defined as those in the bottom nine deciles of the pre-Covid debt distribution among small businesses; large firms are in the top decile. High-contact industries include corner stores, restaurants and bars, hotels, bus passenger, clothing stores, and others as defined in Cerda et al. 2023. All figures represent shares within each lender type.

Traditional banks’ portfolios are more concentrated among large borrowers, who represent 74 percent of bank small-firm credit before the recession, compared to 50 percent for MFIs. This specialization carries over to guarantee allocation: 71 percent of traditional bank guarantees went to large, low-sensitivity borrowers, while MFIs directed 65 percent of their guarantees toward smaller, highly sensitive borrowers.

Similarly, we observe that traditional banks provided more guarantees to low-contact industries, while MFIs attended mainly high-contact sectors. These allocation patterns indicate that MFIs were more likely to provide guarantees to high-sensitive borrowers such as smaller firms in high-contact industries, mirroring their pre-recession specialization.

We conduct a back-of-the-envelope calculation, explained in more detail in Appendix C, to measure the gains from observed MFI participation using our reduced-form evidence. Given our estimates and the observed participation of MFIs and traditional lenders, the program reduced delinquency by 12 percentage points. In contrast, if all guarantees had been allocated through traditional banks, aggregate delinquency would have declined by only 8 percentage points.

6 Quantitative Analysis

We extend our conceptual framework in two directions to perform counterfactual analysis. First, we endogenize firm heterogeneity: a continuum of firms choose their type (soft- or hard-information) and optimal level of debt-financed capital. Second, we include two lenders, traditional banks and MFIs, with different comparative advantages across firm types.

6.1 Firms

We model the timing of firm decisions following Ulyssea (2018). First, each firm receives a private signal ν about its future productivity. This signal is drawn from a Pareto distribution

with shape parameter ξ over the support $[\nu_0, \infty)$, where we normalize $\nu_0 = 1$. Based on this signal, the firm chooses its type, $f \in \{s, h\}$. It can pay a fixed cost c_e to become a hard-information firm ($f = h$), or operate at zero cost as an opaque, soft-information firm ($f = s$). After this choice, the firm's actual productivity z is realized as $z = \epsilon \times \nu$, where ϵ is an idiosyncratic shock drawn from a log-normal distribution with mean zero and variance V .

Firms also face an idiosyncratic probability of failure σ , drawn from a Beta distribution $\sigma \sim \text{Beta}(\alpha_\sigma, \beta_\sigma)$ over $[0, 0.5]$.¹¹ Firms produce using a Cobb-Douglas technology $y = zk^\alpha$, where k denotes capital. After choosing their type f and observing their productivity z and failure risk σ , firms choose their capital level, fully financed with debt, by maximizing profits taking the interest rate r as given, yielding the following capital demand function:

$$k = \left(\frac{\alpha z}{r} \right)^{\frac{1}{1-\alpha}} \quad (7)$$

6.2 Banking Sector

The banking sector includes two lender types: a representative traditional bank and a representative microfinance institution (MFI), each employing distinct lending technologies and competing à la Cournot. This technological difference is captured by their marginal costs of loan origination. The traditional bank has a comparative advantage in serving hard-information firms, reflected in a lower marginal cost $c_h^T < c_h^M$ in that segment. Conversely, the MFI specializes in relationship lending, granting it a cost advantage with opaque, soft-information firms ($c_s^M < c_s^T$). The traditional bank faces higher costs in the soft-information segment due to its reliance on standardized, hard-information techniques. Taking the com-

¹¹We can also relate the risk distribution to firm type, capturing the benefits of information and the inherent vulnerabilities of opaque firms. However, as we documented in previous sections, delinquency rates are similar across different lender types.

petitor's quantity as given, lender j chooses loan supply k_j to maximize:

$$\max_{k_j} (1 - \sigma)r(k)k_j - c_f^j k_j$$

where $k = \sum_j k_j$ is the total credit extended to the firm. The inverse demand function for loans, derived from the firm's optimal capital choice, is $r(k) = \alpha z k^{\alpha-1}$. Then, Lender j 's first order condition is:

$$(1 - \sigma) \left[r + k_j \frac{\partial r}{\partial k} \right] = c_f^j$$

Let $s_j = k_j/k$ denote lender j 's market share in lending to a given firm. In a symmetric equilibrium, where all lenders charge the same interest rate r to a firm of type f with risk σ , we obtain the following equilibrium interest rate:

$$r(\sigma; f) = \frac{c_f}{(1 - \sigma) [1 + (\alpha - 1)\text{HHI}_f]} \quad (8)$$

where $c_f = \sum_j c_f^j \times s_j$ is the weighted average marginal cost and $\text{HHI}_f = \sum_j s_j^2$ is the Herfindahl-Hirschman Index for the segment of firms (σ, f) . This expression captures how firm risk, technological specialization, and lender competition shape credit costs. The rate increases with the firm's default risk σ and the segment's average origination cost c_f , and decreases with market competition (i.e., lower HHI_f). Firms formulate expectations over this pricing schedule when deciding whether to operate as soft- or hard-information firms.

6.3 Equilibrium

After observing their productivity signal ν , firms choose their type by maximizing expected profits. A firm becomes hard-information if the benefit of better financing terms outweighs

the fixed cost c_e :

$$\max_{\{s,h\}} \left\{ \int_{\epsilon,\sigma} \Lambda_s ((1-\sigma)\epsilon\nu)^{\frac{1}{1-\alpha}} dF(\epsilon,\sigma), \int_{\epsilon,\sigma} \Lambda_h ((1-\sigma)\epsilon\nu)^{\frac{1}{1-\alpha}} dF(\epsilon,\sigma) - c_e \right\}$$

where $\Lambda_f = (1-\alpha)(\alpha[1+(\alpha-1)\text{HHI}_f]/c_f)^{\frac{\alpha}{1-\alpha}}$ is a profit shifter that incorporates the equilibrium lending terms for type f , and F denotes the joint distribution of (ϵ, σ) . This problem yields a cutoff equilibrium: firms with signals $\nu \geq \nu^*$ pay c_e and become hard-information, while those with $\nu < \nu^*$ remain opaque, soft-information firms. This endogenous sorting based on expected productivity generates overlapping productivity distributions across the two segments, a key feature for the counterfactual analysis below.

6.4 Loan Guarantee Program

We model the recession as an unexpected liquidity shock and the program as a liquidity injection, as described in the conceptual framework. The total volume of guaranteed loans is M , with an exogenous fraction γ_M channeled through MFIs. Lender $j \in \{\text{TB}, \text{MFI}\}$ solves:

$$\max_{\ell_i^j \in \{0,1\}} \int \ell_i^j \Omega_i^j k_i dG^j(y_i, k_i) \quad \text{s.t.:} \quad \int \ell_i^j \varphi k_i dG^j(y_i, k_i) \leq \gamma_j M \quad (9)$$

Thus, the allocation's efficiency hinges on lender-specific client distributions G^j and budget allocations γ_j . The social planner, by contrast, faces the aggregate firm distribution $G(y, k)$ and maximizes the aggregate treatment effect, following the same objective as in Section 4.

6.5 Calibration

Our calibration is structured in two blocks, each targeting distinct empirical moments. The first block calibrates parameters governing the pre-recession credit market equilibrium. We

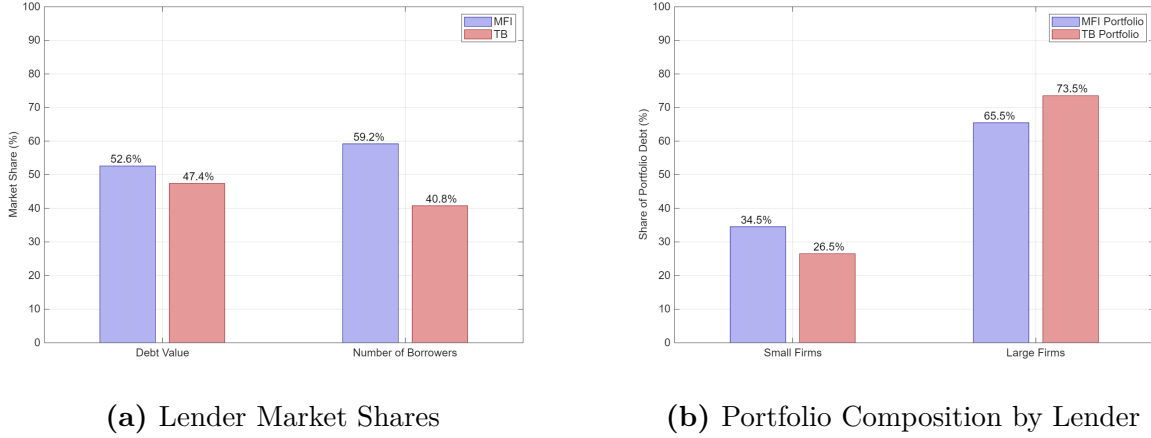
set the capital share $\alpha = 0.6$ to match the aggregate debt-to-sales ratio; c_e replicates the share of informal borrowers (proxied by the tax category “natural person with business”). The Pareto signal parameters (ξ and the upper bound) are set to match the share of total debt held by large borrowers, while the risk-distribution parameters α_σ and β_σ target the average delinquency rate. The lender cost parameters $\{c_f^j\}$ match the average interest rates paid by informal versus formal borrowers. Finally, we set the recovery rate $\delta = 0.1$ based on regulatory estimates, and the relationship value $\psi_F = 2.5$, consistent with a 10% profit margin on a perpetuity discounted at 4%. These parameters and their targets are summarized in the top panel of Table A11 in the Appendix.

Second, we calibrate the parameters governing the loan guarantee program and the recession shock. The total program size M is set to 25% of aggregate outstanding debt; γ_{MFI} is set to 20% to match observed MFI participation. The poaching probability ψ_C is calibrated to 0.1, matching the OLS coefficient of lender switching on guaranteed loan receipt. The credit injection parameter φ replicates the estimated average debt increase among program recipients. Finally, we parameterize the liquidity shock distribution $\tilde{\Phi}(\nu; \eta) = (\nu/c_0)^\eta$ over $[0, c_0]$, where η controls shock severity, jointly calibrating c_0 and η to match our empirical treatment effect estimates by firm size. The bottom panel of Table A11 summarizes these program-related parameters and their targets.

Figure 6 evaluates how well the calibrated model matches key stylized facts of the Peruvian credit market. Panel (a) displays market shares by lender type. Consistent with the descriptive statistics in Section 2, MFIs account for approximately 53% of total debt and, given their specialization in smaller clients, serve about 60% of all borrowers. Panel (b) shows portfolio composition by firm size, where small firms are defined as those in the bottom nine deciles of the debt distribution—similar to our empirical classification. The model successfully reproduces the lender specialization documented earlier: traditional banks allocate 74% of their lending to large firms, whereas MFIs allocate only 66% of their portfolio

to this segment, reflecting their greater focus on smaller borrowers.

Figure 6: Model Equilibrium Market Shares by Lender and Firm-Size



This figure shows the equilibrium market shares of traditional banks and micro-finance institutions in our model. Panel (a) plots the market shares of micro-finance institutions (MFIs) and traditional banks (TBs) in the segment of small firm lending. The left-hand side bars measure market shares in terms of outstanding debt value and the right-hand side bars measure market shares in terms of number of borrowers. The blue bars refer to MFIs and the red bars refer to TBs. Panel (b) plots the market shares of MFIs and TBs by firm-size. Small firms are defined as those at the bottom 90 percent of the debt distribution and large firms are those at the top 10 percent. The blue bars refer to MFIs and the red bars refer to TBs.

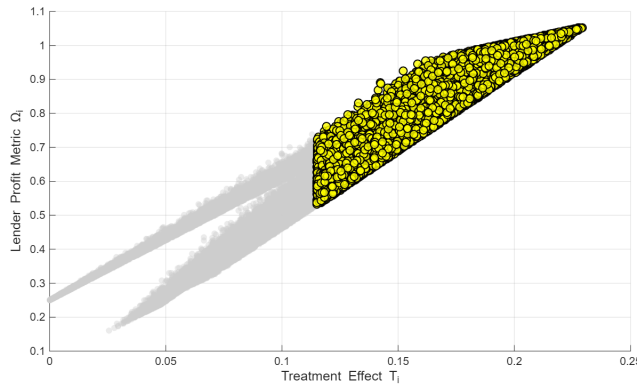
Figure A7 in the Appendix presents equilibrium outcomes by firm size. Panel (a) displays the distribution of net cash-in-hand, defined as $y - rk$, separately for small (blue bars) and large (red bars) firms. Two key patterns emerge. First, small firms are concentrated at lower net-cash values, making them more vulnerable to liquidity shocks. The effect of loan guarantees is non-linear over this distribution: firms at the very bottom may fail even with a guarantee, while those in the middle experience the largest survival probability gains. Second, large firms are predominantly located at higher net-cash levels, implying they are more likely to survive a shock even without government support. Panel (b) plots the average treatment effect T_i by size group. Consistent with our empirical estimates, the treatment effect is approximately four times bigger for smaller firms.

6.6 Equilibrium and Counterfactual Analysis

We use our calibrated model to quantify how MFIs shape the allocation and aggregate effectiveness of loan guarantees during recessions. We begin by comparing the social planner’s guarantee allocation with that in the decentralized market equilibrium. We then compute the optimal MFI participation rate, defined as the share that minimizes the proportion of debt in default.

Allocation of Guarantees. We compare the allocation of guarantees implemented by the social planner with that emerging in the decentralized market equilibrium. Figure 7 illustrates the social planner’s allocation by plotting firms in the space of two characteristics: the lender’s private profit metric Ω_i and the treatment effect T_i . Because the social planner maximizes the total value of non-defaulting debt, the optimal allocation follows a simple priority rule: rank firms by T_i and allocate guarantees in descending order until the budget constraint binds.

Figure 7: Social Planner Allocation of Guarantees



This figure plots the distribution of lender profit metric Ω and treatment effect T across all borrowers. Each dot represents one borrower and yellow dots are those receiving a guarantee from the social planner.

Figure A8 in the Appendix plots the decentralized allocation and illustrates the key trade-off faced by private lenders. The left panel shows the allocation by traditional banks,

who distribute 75% of the guarantee funds. Banks allocate guarantees by ranking clients according to the private profit metric Ω , selecting those with the highest values. Red dots represent firms that receive a guarantee; gray dots represent those that do not. The right panel shows the corresponding allocation by MFIs, who follow the same Ω -based ranking rule but with a smaller budget share (25%), reach a more limited set of clients. The figure illustrates that reallocating a portion of the guarantee budget from traditional banks to MFIs would shift guarantees toward clients with higher treatment effects T , thereby increasing the program’s overall effectiveness.

Optimal MFIs’ Participation. We conduct a counterfactual analysis that varies the share of guarantees distributed by MFIs to identify their optimal participation rate. We define the effectiveness of the program, Δ , as the share of debt at risk of default that is saved by the program:

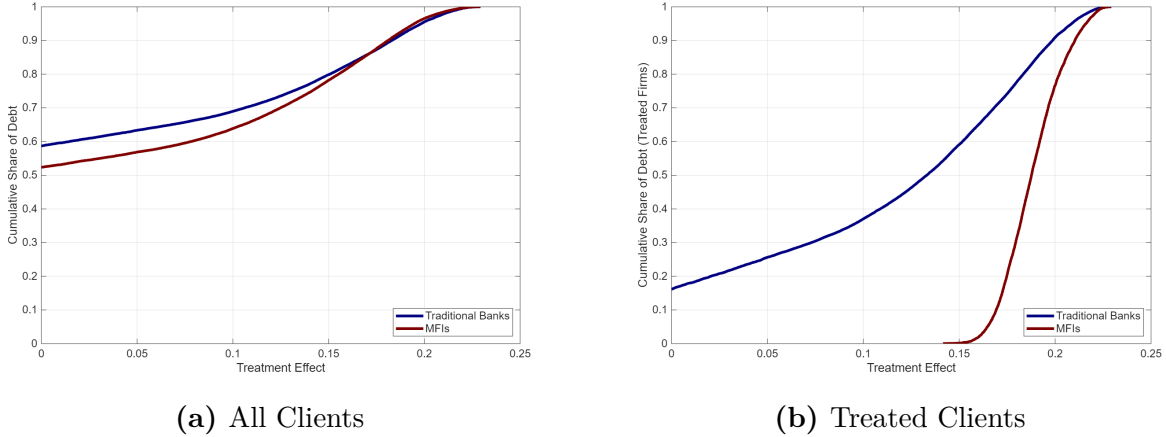
$$\Delta \equiv \frac{\int \ell_i^{\text{TB}}(\Phi_i(1) - \Phi_i(0))(1 - \sigma_i)b_i di + \int \ell_i^{\text{MFI}}(\Phi_i(1) - \Phi_i(0))(1 - \sigma_i)b_i di}{\int (1 - \Phi_i(0))(1 - \sigma_i)b_i di} \quad (10)$$

To understand what drives the optimal MFI participation rate, we first examine how treatment effects vary across each lender’s client pool. Panel (a) of Figure 8 plots the debt-weighted cumulative distribution of treatment effects for traditional bank clients (dashed blue line) and MFI clients (solid red line), and shows that a larger share of traditional banks’ portfolio consists of low-treatment-effect firms. This implies that banks face greater distortionary pressure toward inframarginal lending—allocating guarantees to firms that would survive even without assistance—compared to MFIs.

Panel (b) restricts the sample to firms that actually received guarantees under the observed MFI participation rate. The marginal MFI client, the last firm to receive a guarantee, has a treatment effect exceeding that of approximately 40% of firms served by traditional

banks. Moreover, about 20% of traditional bank clients are inframarginal, with treatment effects near zero. These patterns illustrate the core mechanism: reallocating guarantee funds from traditional banks to MFIs shifts resources toward firms with higher marginal benefits, thereby increasing the program’s aggregate effectiveness.

Figure 8: Distribution of Treatment Effect by Lender Type



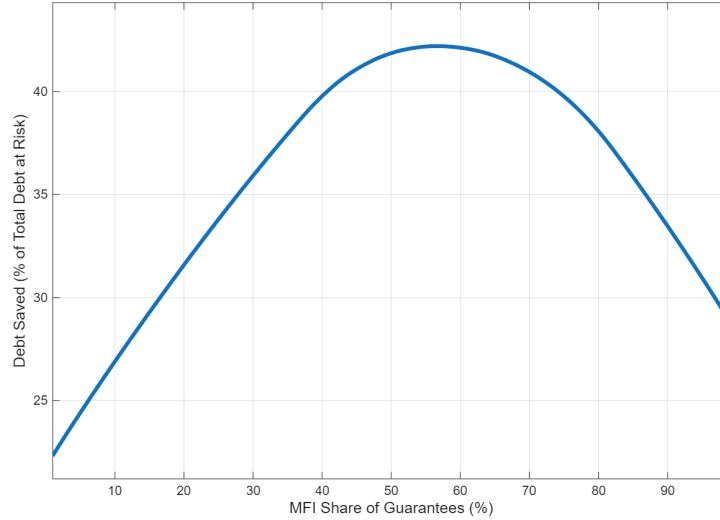
This figure shows the distribution of treatment effects across lender clienteles. Panel (a) plots the debt-weighted cumulative distribution of treatment effects T_i for all clients of traditional banks (dashed blue line) and MFIs (solid red line). Panel (b) restricts the sample to firms that received guarantees under the observed MFI participation rate of 25%.

Figure 9 plots the program effectiveness Δ against the MFI participation rate γ_{MFI} . The function peaks at $\gamma_{\text{MFI}} = 54\%$, where the program saves approximately 45% of debt at-risk. In contrast, exclusive reliance on traditional banks saves only 22%, while exclusive reliance on MFIs saves 28%. The observed equilibrium ($\gamma_{\text{MFI}} = 0.25$) saves 34%—a substantial improvement over the bank-only counterfactual, but still below the optimum.

Two opposing forces shape this relationship. First, MFIs’ comparative advantage in relationship lending means their client pool contains more high-treatment-effect firms. Consequently, at the observed participation rate, a marginal dollar through an MFI reaches a higher-treatment-effect firm than a marginal dollar through a traditional bank, whose portfolio tilts toward inframarginal borrowers. Second, as MFIs distribute more guarantees, they

must serve firms with progressively lower treatment effects, diluting their marginal effectiveness. The optimal participation rate balances these two margins.

Figure 9: Debt Saved by the Program and MFI participation



This figure plots the debt-saved ratio defined by equation (10) for different levels of MFI's participation. This ratio is equal to the total debt saved by the program in the decentralized equilibrium divided by the total debt in default absent the program.

Our model yields three key insights. First, the decentralized allocation deviates from the constrained first-best because private lenders overweight firms' survival probability in the absence of the program. Second, incorporating MFIs mitigates this distortion: their specialization in smaller, opaque firms channels guarantees toward more responsive borrowers. Third, optimal program design requires balancing the benefits of MFI specialization against the costs of extending guarantees to firms that would not survive even with assistance.

6.7 Extensions

Our quantitative findings are informative for settings where MFIs exhibit low delinquency rates and can distribute guarantees as efficiently as traditional banks. Since these charac-

teristics vary across contexts, we extend our model in three directions that capture relevant dimensions of cross-country heterogeneity in MFI development.

First, we introduce an additional cost for MFIs when distributing loan guarantees, capturing constraints such as limited digital infrastructure or scalability that may hinder their effectiveness. Second, we allow MFIs to specialize in riskier borrowers relative to traditional banks. Third, we incorporate lender monitoring incentives that weaken when loans are guaranteed by the government.

MFIs inefficiency in distributing loan guarantees. A potential concern with channeling guarantees through MFIs is that their less sophisticated infrastructure may lead to higher distribution costs, potentially dampening the program’s aggregate effectiveness. To capture this, we introduce a parameter $\tau \in [0, 1]$ that measures the inefficiency of MFI distribution. When $\tau = 0.5$, for example, every *Sol* of guaranteed credit reaching a firm requires the government to provide 1.5 *Soles* to the MFI. We then compute the optimal MFI participation rate for each level of inefficiency.

This extension speaks directly to a key feature of our setting: higher distribution costs led to uncompetitive interest rate bids, limiting MFI participation at the beginning. In response, the Peruvian government launched MFI-specific auctions, offering guaranteed credit at interest rates above those prevailing for traditional banks. By mapping inefficiency to optimal participation, our analysis reveals a key trade-off: expanding MFI participation may improve allocation but raises program expenditure, allowing us to quantify how distribution costs shape optimal program design.

Higher distribution costs reduce the effective loan volume that reaches firms for a given government expenditure through MFIs. This creates a tension in optimal program design: redirecting guarantees toward traditional banks increases the total value of credit effectively distributed—since banks face lower costs—but at the expense of moving funds away from

MFI clients, who tend to have higher treatment effects. The optimal MFI participation rate balances these margins, trading off the efficiency gain from larger effective loan volume against the loss from worse targeting.

To illustrate how distribution costs reshape optimal MFI participation, we start from the frictionless optimum and consider a marginal reallocation: we shift 1% of the guarantee budget from MFIs to traditional banks. This reallocation has two effects. First, some former MFI clients—specifically, those with the lowest Ω among treated firms—lose access to guarantees, reducing the total debt saved from default. Second, new traditional bank clients gain access, increasing debt saved.

Panel (a) of Figure A9 in the Appendix plots these gains and losses across different levels of MFI distribution costs. The gains from reallocating guarantees to traditional banks are constant, as they depend only on the treatment effects of the marginal bank clients. The losses from withdrawing guarantees from MFI clients, however, decline with inefficiency. At low inefficiency levels, below 25%, the losses exceed the gains, making it optimal to maintain MFI participation. Once inefficiency rises above 25%, the losses become smaller than the gains, tilting the optimal allocation toward traditional banks.

Larger losses from a marginal reduction in MFI participation do not imply that increasing their share would be beneficial. The relevant test for an expansion requires comparing the marginal benefit of directing more guarantees through MFIs against the marginal cost of shifting those guarantees away from traditional banks. Panel (b) of Figure A9 plots the optimal MFI share across inefficiency levels. For inefficiency below 25%, the optimal share remains constant at the frictionless level. In this region, the marginal gains from reallocating guarantees toward MFIs are offset by the distribution costs they incur, while the marginal gains from reallocating toward traditional banks are offset by their poorer targeting of high-treatment-effect firms. Above 25%, rising inefficiency makes further MFI participation increasingly costly, and the optimal share declines monotonically. Given the observed

interest rate differential on MFI guarantees relative to traditional banks, the optimal MFI participation rate remains within the constant region.

MFIs specialization in risky borrowers. In our baseline equilibrium, MFI and traditional bank clients exhibit similar risk, consistent with the Peruvian financial sector. However, in other contexts, MFIs may serve riskier borrowers than traditional banks, reducing their optimal participation in guarantee programs because their clients are more likely to default regardless of whether they receive guarantees.

We extend our framework to capture these differences in risk across lenders. We depart from the baseline assumption that the risk distributions of soft- and hard-information firms are equivalent and instead assume that soft-information firms are riskier. We do so by shifting the Beta distribution parameter α_s from 2 to 8 while holding the sum $\alpha_s + \beta_s$ constant, generating variation in the risk profile of soft-information clients. For example, $\alpha_s = 2$ implies an expected default probability of 0.1 for soft-information firms, while $\alpha_s = 8$ raises this expectation to 0.4.

Panel (a) of Figure A10 in the Appendix plots the average risk of MFI clients for different levels of soft-information risk. Given their specialization in soft-information firms, the average default probability of MFI clients rises from 0.10 to 0.28 as soft-information firms become riskier. Consequently, the optimal MFI participation rate declines from 56% to 49%, as shown in Panel (b). This is because riskier clients are more likely to default regardless of guarantee receipt, reducing the marginal benefit of increasing MFI participation.

Monitoring incentives. A potential concern with government guarantees is that they may reduce lenders incentives to monitor borrowers. To explore this channel, we extend our framework to account for bank monitoring. For simplicity, we assume that the equilibrium firm debt obligations and lender attachment is given by our baseline equilibrium described

above. In the online appendix, we provide a full extension where the initial pre-recession equilibrium also incorporates bank monitoring decisions and standard Cournot competition.

The pre-recession equilibrium includes each firm of type z matched to bank j with a debt obligation $Rk_j(z)$. We depart in three directions. First, we consider that firms can divert some of the capital and that lenders can monitor firms to reduce diversion. Thus, the effective capital used by firms in production is $k_j^{\text{eff}} = (\theta + (1 - \theta)m_j)k_j$, where $1 - \theta$ is the fraction of wasted capital absent monitoring, and m_j is bank j monitoring effort. Second, firm revenue is distorted by an idiosyncratic shock ε , drawn from a log-normal distribution $H(\varepsilon)$, such that $y = z\varepsilon \ln(k_j^{\text{eff}})$. Firms face limited liability and default if the realized output is below debt obligations, even absent the recession. Third, banks can recover a fraction $1 - \lambda$ of realized output when default occurs.

The recession is a lump-sum shock ν that reduces firm revenue to $y = z\varepsilon \ln(k_j^{\text{eff}}) - \nu$. The bank only observes the distribution $\Phi(\nu)$ and adjusts monitoring before the shock and firm idiosyncratic productivity are realized. Thus, default occurs when:

$$z\varepsilon \ln(k_j^{\text{eff}}) < Rk_j + \nu \quad (11)$$

Let $\tilde{m}_j(z)$ be the optimal monitoring and $\tilde{\mu}_j$ the corresponding fraction of capital that is not diverted $\tilde{\mu}_j \equiv \theta + (1 - \theta)\tilde{m}_j(z)$. For a given firm type z , expected recovery is given by:

$$\bar{\mathcal{R}}_j(z, \tilde{m}) = \int_0^{\nu_0} (1 - \lambda) \left[z(\ln k_j + \ln \tilde{\mu}_j) \left(I(\varepsilon_j^{*,\nu}) - I(\varepsilon_j^{**}) \right) - \nu \left(H(\varepsilon_j^{*,\nu}) - H(\varepsilon_j^{**}) \right) \right] d\Phi(\nu)$$

where $\varepsilon_j^{*,\nu}$ is the default threshold below which firms cannot repay the bank, ε_j^{**} is the zero-recovery threshold below which output does not cover the shock, and $I(a) \equiv \int_0^a \varepsilon h(\varepsilon) d\varepsilon$.

When the recession hits, the bank chooses $\tilde{m}_j(z)$ to maximize expected profit from type- z

firm with debt obligations Rk_j , excluding funding costs:

$$\tilde{\Pi}_j(z, \tilde{m}) = [1 - \bar{\pi}_j(z, \tilde{m})] Rk_j + \bar{\mathcal{R}}_j(z, \tilde{m}) - \frac{c_j(z)}{2} \tilde{m}^2 k_j$$

where $\bar{\pi}_j(z, \tilde{m})$ denotes the expected default probability and $c_j m^2/2$ is the monitoring cost per unit of capital. The first order condition on monitoring is:

$$\int_0^{\nu_0} \left[h(\varepsilon_j^{*,\nu}) \left| \frac{\partial \varepsilon_j^{*,\nu}}{\partial \tilde{m}} \right| \lambda Rk_j + (1 - \lambda) z \frac{(1 - \theta)}{\tilde{\mu}_j} (I(\varepsilon_j^{*,\nu}) - I(\varepsilon_j^{**})) \right] f_\nu(\nu) d\nu = c_j(z) \tilde{m}_j^u k_j \quad (12)$$

This equation determines the optimal monitoring in the recession without government intervention $\tilde{m}_j^u(z)$. For each realization ν , the derivative of per-firm profit with respect to $\tilde{m}$ has two components. At the extensive margin, the bank recovers $(1 - \lambda)Rk_j$ from the marginal defaulter. Such recovery is independent of ν . At the intensive margin, higher monitoring raises effective capital, increasing output for all defaulting firms in the recovery region $[\varepsilon_j^{**}, \varepsilon_j^{*,\nu}]$.

With government guarantees, the first order condition on monitoring follows the same structure as (12) but with modified thresholds $\varepsilon_j^{*,\nu,s}$ and $\varepsilon_j^{**,s}$. The impact of government guarantees on optimal monitoring is twofold. First, guarantees reduce the sensitivity of the threshold to monitoring. Second, the recovery region for supported firms $[\varepsilon_j^{**,s}, \varepsilon_j^{*,\nu,s}]$ has a lower upper boundary, reducing $I(\varepsilon_j^{*,\nu,s}) - I(\varepsilon_j^{**,s})$. Since the monitoring cost function is unchanged, the optimal monitoring among supported firms $\tilde{m}_j^s$ is lower than that for unsupported ones $\tilde{m}_j^u$.

Conditional on optimal monitoring, bank j obtains the following gain from providing guaranteed credit to firm z :

$$\Delta \Pi_j(z) = \tilde{\Pi}_j^s(z) - \tilde{\Pi}_j^u(z) \quad (13)$$

This captures the full effect of providing loan guarantees, including the endogenous monitoring adjustment. Bank j ranks all firms by $\omega_j(z) = \Delta \Pi_j(z) / \varphi k_j(z)$ and supports type z if $\omega_j(z) \geq \bar{\omega}_j$, where the cutoff is determined by the budget constraint.

The effectiveness of the program declines to the extent that lenders replace monitoring effort with government guarantees. Since monitoring is more important for smaller borrowers, banks rationally devote less effort to supporting them when those firms are shielded from default by government guarantees. As a result, MFIs, which are most effective at serving small, opaque borrowers, experience the largest drop in monitoring for their core clients, reducing the program's aggregate effectiveness and lowering the optimal MFI participation rate.

7 Conclusions

The global expansion of microfinance institutions in emerging markets has raised important questions about their capacity to support long-term economic development and facilitate short-term recovery during economic crises. Using a theoretical framework and a large-scale loan guarantee program in Peru as our empirical setting, we provide evidence on how MFIs shape the effectiveness of financial stimulus during economic downturns in developing countries.

We show that loan guarantees significantly increased credit supply and improved firm performance, with stronger effects among smaller firms and those in high-contact industries, the segments facing the most severe liquidity needs. MFIs played a central role in channeling guarantees to these high-sensitivity segments, reflecting their pre-recession specialization in relationship-based lending. We develop and calibrate a model in which lenders specialize in different client types, finding that distributing guarantees solely through traditional banks would have saved only 22 percent of debt at risk of default, whereas the actual allocation

with MFI participation raised this ratio to 34 percent.

Our results have important policy implications. MFIs are valuable policy actors, particularly for reaching small entrepreneurs who are most responsive to credit but often underserved by traditional banks. However, our findings caution against over-reliance on MFIs: at high participation rates, MFIs would inevitably serve clients with lower treatment effects, reducing the aggregate effectiveness of guarantees.

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Appendix A: Additional Tables

Table A1: Borrower Characteristics by Loan Type

	<u>Total Loans</u>		<u>Repayment</u>	<u>Number</u>
	Mean	Median	<u>Delay</u>	<u>of firms</u>
	(1)	(2)	(3)	(4)
Total	6	0.5	0.12	2 854
<i>Loans to:</i>				
Micro-credit	1	0.5	0.10	2 290
Small firms	11	7	0.14	545
Medium-size firms	116	30	0.23	36
Large firms	690	85	0.10	3
Corporations	5 850	630	0.03	0.5

This table reports firm-level summary statistics as of December 2019. We report the mean and median of the distribution of total loans across firms. Repayment delay denotes the share of firms exhibiting more than 30 days of repayment delay. Total loans are expressed in USD thousand, and number of firms is expressed in thousand.

Table A2: Traditional Banks and Micro-Finance Institutions

	<u>Traditional Banks</u>		<u>Micro-Finance Inst.</u>		<u>Top 10 MFIs</u>	
	Mean	Median	Mean	Median	Mean	Median
	(1)	(2)	(3)	(4)	(5)	(6)
Total Assets	7.89	1.78	0.39	0.16	1.04	0.98
Total Credit	5.45	1.25	0.33	0.16	0.94	0.86
Delinquency Rate	3.57	3.02	7.81	5.46	6.18	4.71
ROA	1.70	2.00	1.64	1.41	1.89	2.23
Num. of Cities	61	46	46	40	99	94
Geographical loan concentration	.48	.32	.21	.05	.03	.03
Num. Institutions	15		37		10	
<i>In small business loans:</i>						
Total Credit	0.42	0.12	0.12	0.05	0.36	0.36
Num. Institutions	10		35		10	

This table reports bank-level summary statistics of institutions participating in the segment of small business loans as of December 2019. Columns (1) and (2) consider all traditional banks, columns (3) and (4) include all microfinance institutions, and columns (5) and (6) focus on the 10 biggest MFIs according to their total value of credit. The value of assets and credit are expressed in USD billion, while delinquency and ROA are expressed as percentages. Geographical concentration for bank b is computed using bank b loans in city c , L_{cb} , as follows: $\sum_c (L_{cb}/L_b)^2$

Table A3: Effect of Loan Guarantees on New Lending Relationships

	<u>Value of New loans</u>		<u>Share of New Loans</u>	
	(1)	(2)	(3)	(4)
Exposure _{<i>b</i>} × Post _{<i>t</i>}	0.434** (0.164)	0.415*** (0.135)	0.023* (0.012)	0.029** (0.013)
Observations	334	334	478	478
Fixed Effects				
Bank	Yes	Yes	Yes	Yes
Quarter	Yes	-	Yes	-
Bank-size × Quarter	No	Yes	No	Yes

This table shows the effect of the program on the balance of total loans and non-Covid-19 loans at the bank-firm level. Treatment is standardized. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively. Standard errors are clustered at the bank level. Observations are expressed in millions.

Table A4: Heterogeneous Effect of Loan Guarantees on Small Firm Financial Performance
- Type of Credit

	Micro-Credit			Small Business Loan		
	OLS	IV		OLS	IV	
	(1)	(2)	(3)	(4)	(5)	(6)
$\text{Exposure}_i \times \text{Post}_t$	-0.021*** (0.004)			-0.009* (0.005)		
$\text{Treated}_i \times \text{Post}_t$		-0.122*** (0.024)			-0.039* (0.021)	
$\ln \text{Loans}_{it}$			-0.169*** (0.041)			-0.042* (0.021)
Observations	19.8M	19.8M	19.8M	4.8M	4.8M	4.8M
Fixed Effects						
Firm	Yes	Yes	Yes	Yes	Yes	Yes
$x^{\text{Real}}(f) \times \text{Quarter}$	Yes	Yes	Yes	Yes	Yes	Yes
$x^{\text{Fin.}}(f) \times \text{Quarter}$	Yes	Yes	Yes	Yes	Yes	Yes

This table shows the effects of being better connected to treated banks on the balance of total loans, non-Covid-19 loans, and delinquency rates at the firm level. Treatment is standardized. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively. Standard errors are clustered at the firm level.

Table A5: Effect of Loan Guarantees on Real Outcomes - Micro Firms

	Capital	Wage-bill	Labor	Sales	Exit in		
					t = 0	t = 1	t = 2
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Exposure _i × Post _t	0.021** (0.008)	0.020 (0.012)	0.012 (0.011)	0.020** (0.009)			
Exposure _i					-0.0113*** (0.0031)	-0.0203*** (0.0036)	-0.0229*** (0.0048)
Observations	586,592	253,763	255,118	595,705	74,729	74,729	74,729
Fixed Effects							
Firm	Yes	Yes	Yes	Yes	-	-	-
Size-bin × Quarter	Yes	Yes	Yes	Yes	-	-	-
Leverage-bin × Quarter	Yes	Yes	Yes	Yes	-	-	-
Industry × Quarter	Yes	Yes	Yes	Yes	-	-	-
City × Quarter	Yes	Yes	Yes	Yes	-	-	-
Size-bin	-	-	-	-	Yes	Yes	Yes
Leverage-bin	-	-	-	-	Yes	Yes	Yes
Industry	-	-	-	-	Yes	Yes	Yes
City	-	-	-	-	Yes	Yes	Yes

This table shows the effect of the program on the balance of total loans and non-Covid-19 loans at the bank-firm level. Treatment is standardized. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively. Standard errors are clustered at the bank level. Observations are expressed in millions.

Table A6: Drivers of Real Effects - Micro Firms

	Δ Capital				Δ Wage-bill			
	> 0	$> p90$	< 0	$< p10$	> 0	$> p90$	< 0	$< p10$
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Exposure _{<i>i</i>}	0.0097*** (0.0020)	0.0000 (0.0017)	-0.0118*** (0.0019)	-0.0185*** (0.0028)	-0.0043 (0.0028)	-0.0119*** (0.0030)	0.0060* (0.0031)	-0.0123* (0.0062)
Observations	73,373	73,373	73,373	73,373	29,314	29,314	29,314	29,314
Fixed Effects								
Size-bin	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Leverage-bin	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
City	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

This table shows the effect of the program on the balance of total loans and non-Covid-19 loans at the bank-firm level. Treatment is standardized. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively. Standard errors are clustered at the industry level. Observations are expressed in millions.

Table A7: Heterogeneous Effect of Loan Guarantees on Small Firm Financial Performance
- Industry Contact in Bottom Deciles of Pre-Covid Debt

	High-Contact			Low and Non-Contact		
	OLS	IV		OLS	IV	
	(1)	(2)	(3)	(4)	(5)	(6)
$\text{Exposure}_i \times \text{Post}_t$	-0.022*** (0.004)			-0.016*** (0.005)		
$\text{Treated}_i \times \text{Post}_t$		-0.130*** (0.021)			-0.099*** (0.031)	
$\ln \text{Loans}_{it}$			-0.182*** (0.036)			-0.101*** (0.032)
Observations	14.7M	14.7M	14.7M	7.0M	7.0M	7.0M
Fixed Effects						
Firm	Yes	Yes	Yes	Yes	Yes	Yes
$x^{\text{Real}}(f) \times \text{Quarter}$	Yes	Yes	Yes	Yes	Yes	Yes
$x^{\text{Fin.}}(f) \times \text{Quarter}$	Yes	Yes	Yes	Yes	Yes	Yes

This table shows the effects of being better connected to treated banks on the balance of total loans, non-Covid-19 loans, and delinquency rates at the firm level. Treatment is standardized. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively. Standard errors are clustered at the firm level.

Table A8: Heterogeneous Effect of Delinquency on Small Firm Financial Performance - Bottom Deciles

Panel A						
	MFI share = 1			MFI share $\in (1, 0.5]$		
	OLS	IV		OLS	IV	
	(1)	(2)	(3)	(4)	(5)	(6)
Exposure _{<i>i</i>} $\times$ Post _{<i>t</i>}	-0.028*** (0.008)			-0.024** (0.009)		
Treated _{<i>i</i>} $\times$ Post _{<i>t</i>}		-0.225** (0.084)			-0.163** (0.065)	
ln Loans _{<i>it</i>}			-0.427** (0.169)			-0.171*** (0.050)
Observations	9.5M	9.5M	9.5M	1.3M	1.3M	1.3M
Panel B						
	MFI share $\in (0.5, 0)$			MFI share = 0		
	OLS	IV		OLS	IV	
Exposure _{<i>i</i>} $\times$ Post _{<i>t</i>}	-0.021*** (0.001)			-0.009*** (0.002)		
Treated _{<i>i</i>} $\times$ Post _{<i>t</i>}		-0.161*** (0.009)			-0.061** (0.021)	
ln Loans _{<i>it</i>}			-0.123*** (0.018)			-0.034*** (0.009)
Observations	1M	1M	1M	2.7M	2.7M	2.7M
Fixed Effects						
Firm	Yes	Yes	Yes	Yes	Yes	Yes
$x^{\text{Real}}(f) \times \text{Quarter}$	Yes	Yes	Yes	Yes	Yes	Yes
$x^{\text{Fin.}}(f) \times \text{Quarter}$	Yes	Yes	Yes	Yes	Yes	Yes

This table shows the effects of being better connected to treated banks on the balance of total loans, non-Covid-19 loans, and delinquency rates at the firm level. Treatment is standardized. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively. Standard errors are clustered at the firm level.

Table A9: Heterogeneous Effect of Delinquency on Small Firm Financial Performance - Micro-credit

Panel A						
	MFI share = 1			MFI share $\in (1, 0.5]$		
	OLS	IV		OLS	IV	
	(1)	(2)	(3)	(4)	(5)	(6)
Exposure _{<i>i</i>} $\times$ Post _{<i>t</i>}	-0.024*** (0.008)			-0.021** (0.009)		
Treated _{<i>i</i>} $\times$ Post _{<i>t</i>}		-0.204** (0.084)			-0.154** (0.068)	
ln Loans _{<i>it</i>}			-0.355** (0.135)			-0.149*** (0.050)
Observations	13.4M	13.4M	13.4M	1.9M	1.9M	1.9M
Panel B						
	MFI share $\in (0.5, 0)$			MFI share = 0		
	OLS	IV		OLS	IV	
	(1)	(2)	(3)	(4)	(5)	(6)
Exposure _{<i>i</i>} $\times$ Post _{<i>t</i>}	-0.022*** (0.001)			-0.009*** (0.002)		
Treated _{<i>i</i>} $\times$ Post _{<i>t</i>}		-0.172*** (0.010)			-0.065** (0.021)	
ln Loans _{<i>it</i>}			-0.135*** (0.016)			-0.035*** (0.009)
Observations	1.5M	1.5M	1.5M	4.6M	4.6M	4.6M
Fixed Effects						
Firm	Yes	Yes	Yes	Yes	Yes	Yes
$x^{\text{Real}}(f) \times \text{Quarter}$	Yes	Yes	Yes	Yes	Yes	Yes
$x^{\text{Fin.}}(f) \times \text{Quarter}$	Yes	Yes	Yes	Yes	Yes	Yes

This table shows the effects of being better connected to treated banks on the balance of total loans, non-Covid-19 loans, and delinquency rates at the firm level. Treatment is standardized. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively. Standard errors are clustered at the firm level.

Table A10: Heterogeneous Effect of Delinquency on Small Firm Financial Performance - High-Contact

Panel A						
	MFI share = 1			MFI share $\in (1, 0.5]$		
	OLS	IV		OLS	IV	
	(1)	(2)	(3)	(4)	(5)	(6)
Exposure _{<i>i</i>} $\times$ Post _{<i>t</i>}	-0.027*** (0.008)			-0.022** (0.010)		
Treated _{<i>i</i>} $\times$ Post _{<i>t</i>}		-0.229** (0.085)			-0.150** (0.067)	
ln Loans _{<i>it</i>}			-0.421** (0.162)			-0.155*** (0.054)
Observations	10.1M	10.1M	10.1M	1.6M	1.6M	1.6M
Panel B						
	MFI share $\in (0.5, 0)$			MFI share = 0		
	OLS	IV		OLS	IV	
	(1)	(2)	(3)	(4)	(5)	(6)
Exposure _{<i>i</i>} $\times$ Post _{<i>t</i>}	-0.010 (0.007)			-0.001 (0.003)		
Treated _{<i>i</i>} $\times$ Post _{<i>t</i>}		-0.057 (0.035)			-0.003 (0.013)	
ln Loans _{<i>it</i>}			-0.055 (0.016)			-0.002 (0.009)
Observations	1.4M	1.4M	1.4M	3.2M	3.2M	3.2M
Fixed Effects						
Firm	Yes	Yes	Yes	Yes	Yes	Yes
$x^{\text{Real}}(f) \times \text{Quarter}$	Yes	Yes	Yes	Yes	Yes	Yes
$x^{\text{Fin.}}(f) \times \text{Quarter}$	Yes	Yes	Yes	Yes	Yes	Yes

This table shows the effects of being better connected to treated banks on the balance of total loans, non-Covid-19 loans, and delinquency rates at the firm level. Treatment is standardized. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively. Standard errors are clustered at the firm level.

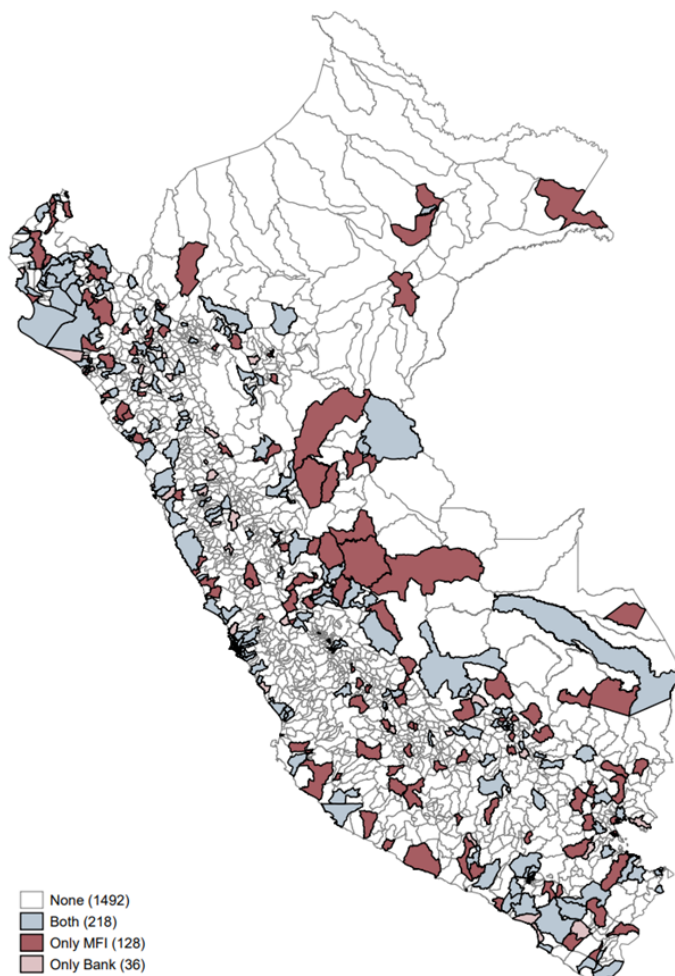
Table A11: Model Calibration

Parameter	Description	Value	Target
α	Capital share in production	0.6	Aggregate leverage ratio
c_e	Fixed cost for hard-info firms	0.045	Share of formal borrowers
$\xi, b_{\max}$	Truncated Pareto signal	2.5, 1.5	Large firm debt share
V	Log-normal productivity variance	0.2	Share of large informal firms
$\alpha_\sigma, \beta_\sigma$	Beta distribution for firm risk	2, 8	Delinquency rate
$c_s^M, c_s^B, c_h^M, c_h^B$	Loan origination costs	1.07, 1.35, 1.07, 1.01	Lender specialization patterns
δ	Recovery rate	0.1	Recovery rate in default
ψ_F	Relationship value	2.5	10% returns and 4% discount
m	Guarantees over aggregate debt	0.25	Observed size of the program
γ_M	Guarantees distributed by MFIs	0.25	Observed MFI participation
ψ_C	Poaching probability	0.1	Unserved switching lender
φ	Credit growth with guarantees	0.8	Debt increase for recipients
c_0, η	Shock distribution	0.5, 0.4	Average treatment effects

This table presents the calibrated parameters for the endogenous banking model with loan guarantees. The top panel shows the parameters that govern the equilibrium distribution of firms across information types (soft vs hard) and lenders (MFIs vs traditional banks). The bottom panel shows the parameters that determine the allocation of government guarantees. Column (2) describes the parameters, column (3) shows the calibrated values, and column (4) outlines the targeted moment.

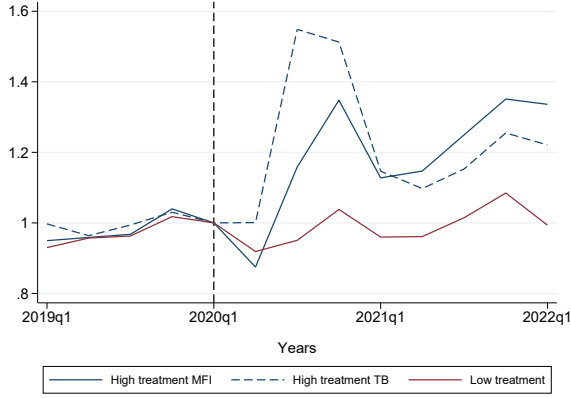
Appendix B: Additional Figures

Figure A1: Geographical Distribution of Financial Institutions

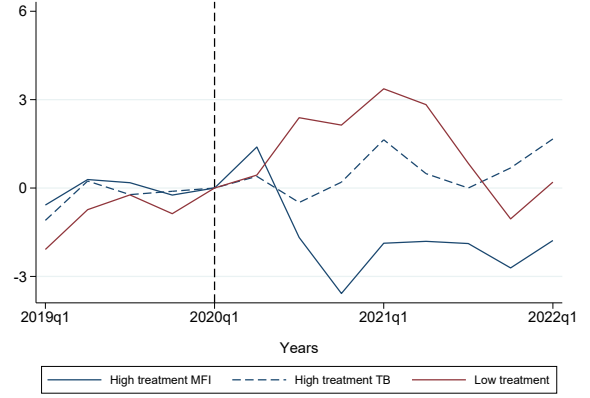


This figure maps the geographical distribution of financial institutions across Peruvian cities as of December 2019. Each city is colored according to the type of lender present: cities with only traditional bank branches, cities with only MFI branches, cities with both, and cities with no financial institution (financial deserts).

Figure A2: Credit, Delinquency and Bank Treatment



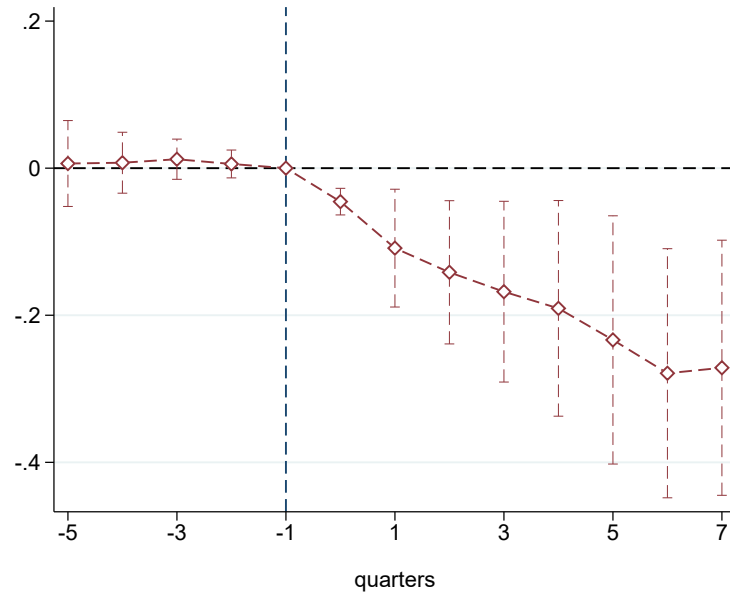
(a) Total Loans



(b) Delinquency Rates

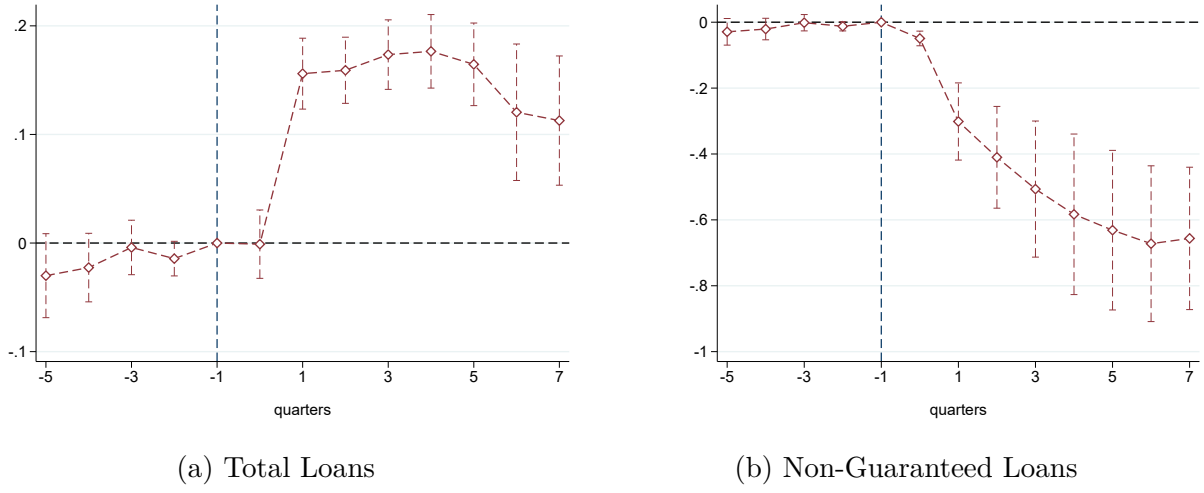
This figure plots the evolution of credit and delinquency rates for high and low treated banks according to our measure of treatment defined by equation (2). Panel (a) plots credit growth rate measured by the value of credit in a given point in time relative to the corresponding value in 2019q4. Panel (b) plots delinquency rates growth measured by the share of outstanding debt with more than 30 days of repayment delay in a given point in time minus the corresponding value in 2019q4. The dashed line corresponds to 2020q1, the quarter prior to the program.

Figure A3: Effect of the Program on Non-Guaranteed Loans



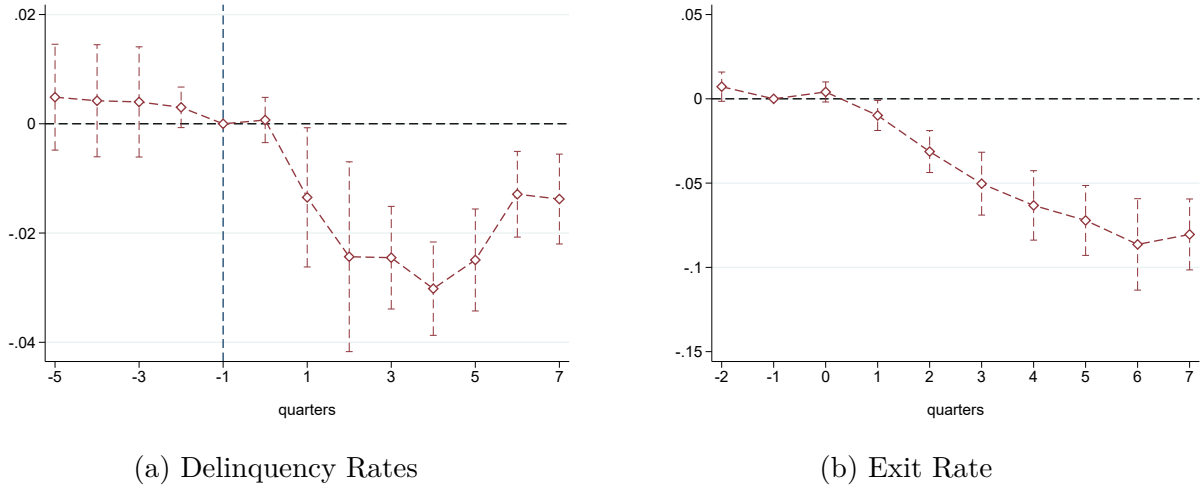
This figure plots the quarterly effects of the program on total credit and non-Covid loans at the bank-firm level. The dependent variable is in logs. The program is implemented in the second quarter of 2020. Each dot is the coefficient on the interaction of treatment and quarter fixed effects. We normalize the treatment effect at the quarter right before the implementation of the program to be equal to zero. Confidence intervals at 95%.

Figure A4: Lending Relationships and Credit



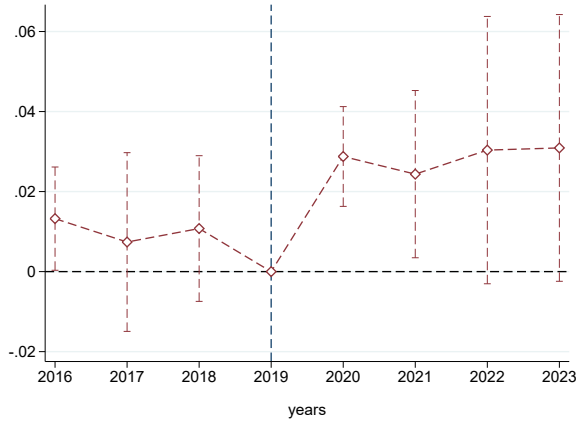
This figure plots the quarterly effects of being better connected to treated banks on total credit and non-Covid-19 loans at the firm level. The dependent variables are in logs. The program is implemented in the second quarter of 2020. Each dot is the coefficient on the interaction of treatment and quarter fixed effects. We normalize the treatment effect at the quarter right before the implementation of the program to be equal to zero. The confidence interval is at the 95% level.

Figure A5: Lending Relationships and Delinquency Rates

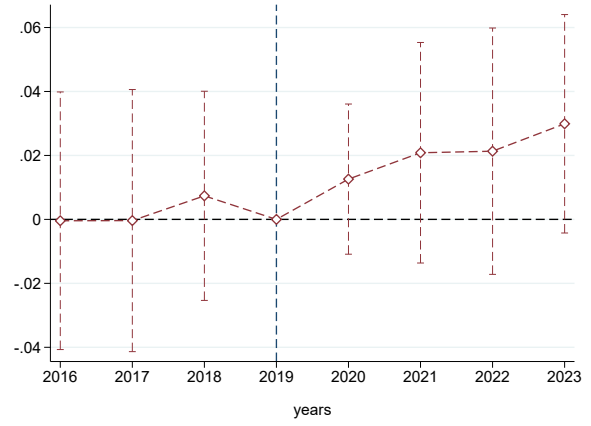


This figure plots the quarterly effects of being better connected to treated banks on delinquency rates, defined as an indicator variable of experiencing repayment delays. The program is implemented in the second quarter of 2020. Each dot is the coefficient on the interaction of treatment and quarter fixed effects. We normalize the treatment effect at the quarter right before the implementation of the program to be equal to zero. The confidence interval is at the 95% level.

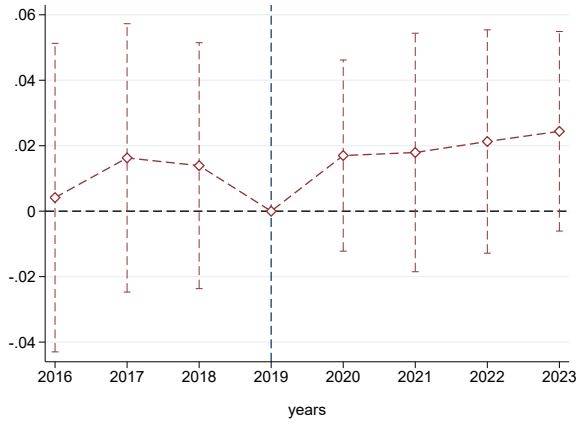
Figure A6: Lending Relationships and Real Outcomes



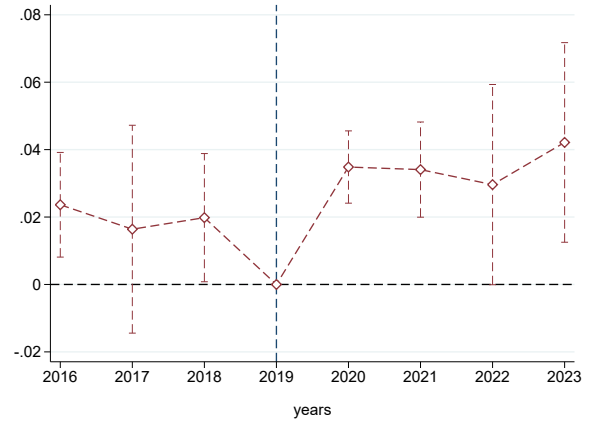
(a) Capital



(b) Wage-bill



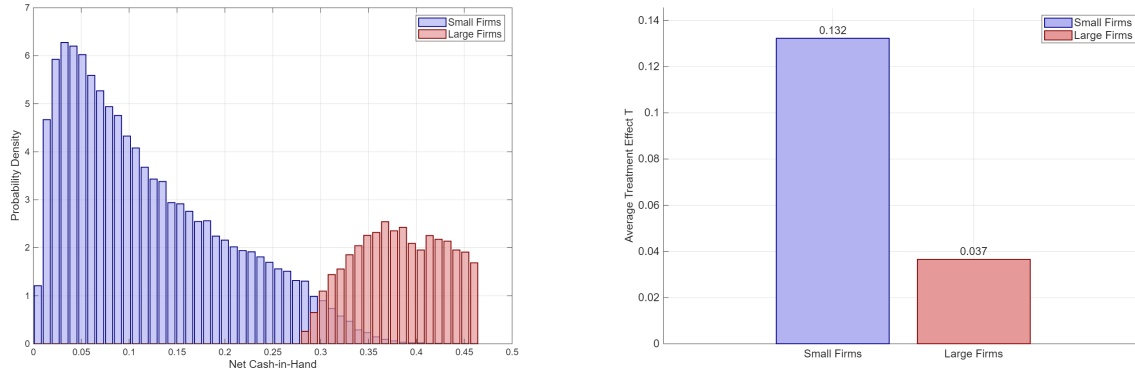
(c) Num. Workers



(d) Sales

This figure plots the annual effects of being better connected to treated banks on delinquency rates, defined as an indicator variable of experiencing repayment delays. The program is implemented in the second quarter of 2020. Each dot is the coefficient on the interaction of treatment and quarter fixed effects. We normalize the treatment effect at the quarter right before the implementation of the program to be equal to zero. The confidence interval is at the 95% level.

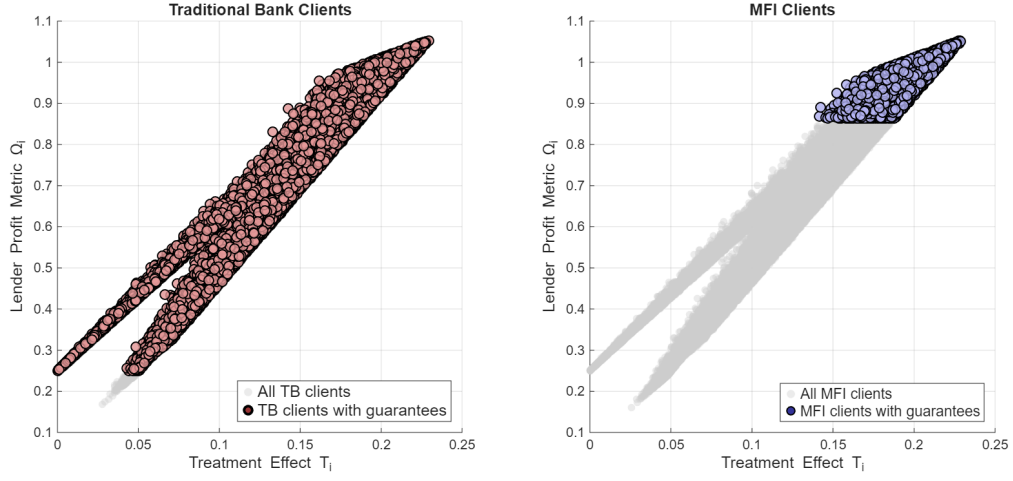
Figure A7: Net Cash Holdings and Treatment Effects by Firm-Size



(a) Net Cash Holding Distribution by Firm-Size **(b)** Average Treatment Effect by Firm Size

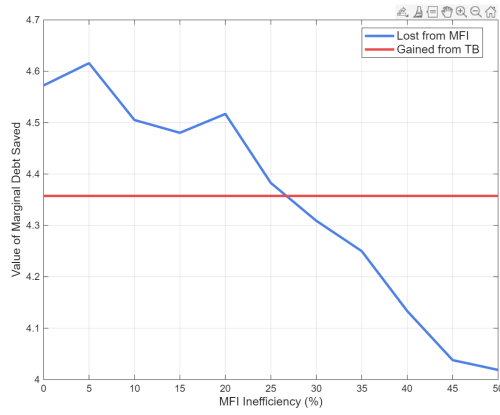
Panel(a) plots the distribution of net cash holdings across firms of different size. Net cash is defined by $y - rk$. Panel (b) plots the average treatment effect T_i by firm-size. Blue bars consider small firms and red bars consider large firms. Small firms are those at the bottom 90 percent of the debt distribution and large firms are at the top 10 percent.

Figure A8: Private Allocation of Guarantees

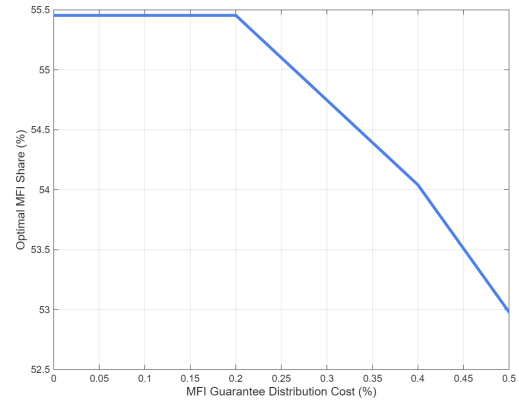


This figure plots the allocation of guarantees by private lenders. Panel (a) shows the distribution of lender profit metric Ω and treatment effect T across traditional banks (TBS) clients. The dots highlighted in red represent firms that receive a loan guarantee from TBs. Panel (b) shows the distribution of lender profit metric Ω and treatment effect T across micro-finance institutions (MFIs) clients. The dots highlighted in blue represent firms that receive a loan guarantee from MFIs.

Figure A9: Distribution Costs and Optimal MFI Participation



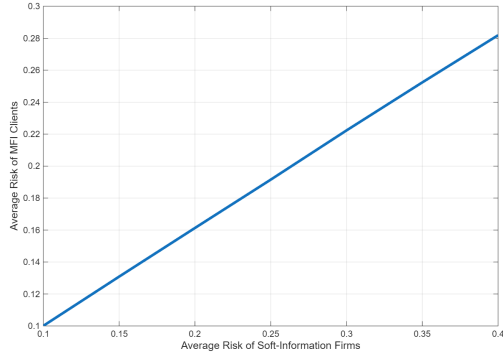
(a) Marginal Gains and Losses from Reallocation



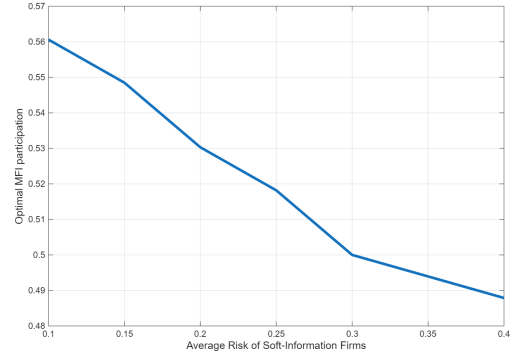
(b) Optimal MFI Participation Rate

This figure shows how distribution costs shape optimal MFI participation. Panel (a) plots the marginal impact of reallocating 1% of guarantees from MFIs to traditional banks for different levels of distribution inefficiency τ . Panel (b) plots the resulting optimal MFI participation rate.

Figure A10: Optimal MFI Participation with Riskier Borrowers



(a) Average Risk of MFI Clients



(b) Optimal MFI Participation Rate

This figure shows how borrower risk shapes optimal MFI participation. Panel (a) plots the average default probability of MFI clients as we shift α_s , making soft-information firms riskier. Panel (b) plots the resulting optimal MFI participation rate. The parameter α_s governs the risk distribution of soft-information firms; higher values imply riskier borrowers. We hold $\alpha_s + \beta_s$ constant to isolate the shift in the distribution's mean. The x-axis reports $\frac{\alpha_s}{\alpha_s + \beta_s} \times 0.5$, which maps to the implied expected default probability of soft-information firms.

Appendix C: Back-of-the-Envelope Calculation

Consider the following stock of debt in period t , before the recession:

$$\overline{B}_t = B_t^n + B_t^d$$

Out of total debt $\overline{B}_t$, the non delinquent debt is B_t^n , while delinquent debt is B_t^d . Then, we have the following delinquent debt in $t + 1$, after the recession, under the assumption that no program was implemented:

$$B_{t+1}^d = \Pr(d|n) B_t^n + \Pr(d|d) B_t^d$$

Where $\Pr(d|n)$ denotes the probability of being delinquent in $t + 1$ conditional on being non delinquent in t , and $\Pr(d|d)$ denotes the probability of being delinquent in $t + 1$ conditional on being delinquent in t .

Since the program was delivered only to non-delinquent borrowers, we assume that only $\Pr(d|n)$ was affected by the program. Thus, we have the following delinquent debt in $t + 1$, under the program:

$$\tilde{B}_{t+1}^d = \Pr(d|n, T) B_t^{n,T} + \Pr(d|n, C) B_t^{n,C} + \Pr(d|d) B_t^d$$

Where $\Pr(d|n, T)$ denotes the probability of being delinquent in $t + 1$ conditional on being non delinquent in t and being treated, and $\Pr(d|n, C)$ denotes the same probability for non-treated borrowers. $B_t^{n,T}$ and $B_t^{n,C}$ represent debt holdings of treated and non-treated firms.

Thus, the program saved the following share of debt from default:¹²

$$\frac{\tilde{B}_{t+1}^d - B_{t+1}^d}{\bar{B}_t} = (\Pr(d|n, T) - \Pr(d|n)) \times \frac{B_t^{n,T}}{\bar{B}_t}$$

Now, we can compute this using our reduced form estimates. Column (1) of Table ?? shows that the impact of credit growth on probability of default is 0.304. Thus, we can define:

$$\Pr(d|n, T) - \Pr(d|n) = -0.304 \times \text{Credit Growth} = -0.304 \times \frac{M}{B^{n,T}}$$

Since guaranteed loans represented 40 percent of current small business loans (see Table 2), i.e., $\frac{M}{B_t} = 0.4$, then:

$$\frac{\tilde{B}_{t+1}^d - B_{t+1}^d}{\bar{B}_t} = -0.304 \times 0.4 = -0.122$$

We can conclude that the program saved 12 percent of existing debt from default.

What if guarantees were distributed by traditional banks only? In this case, the elasticity would have been different. Columns (2) and (3) of Table 6 show that the impact of credit growth on default probability is 0.463 among smaller borrowers and 0.164 among larger ones. Since traditional banks distribute 18 percent of guarantees to the former and 82 percent to the latter, we have:

$$\Pr(d|n, T) - \Pr(d|n) = -(0.463 \times 0.18 + 0.164 \times 0.82) \times \text{Credit Growth} = -0.218 \times \frac{M}{B^{n,T}}$$

Finally, we have:

$$\frac{\tilde{B}_{t+1}^d - B_{t+1}^d}{\bar{B}_t} = -0.087$$

¹²Of course, we are implicitly assuming that $\Pr(d|n) = \Pr(d|n, C)$ and $\Pr(d|d)$ does not change because of the program, i.e., we are abstracting from any type of general equilibrium effects.

which means that the program would have saved 8.7 percent of existing debt if all guarantees were distributed by traditional banks.